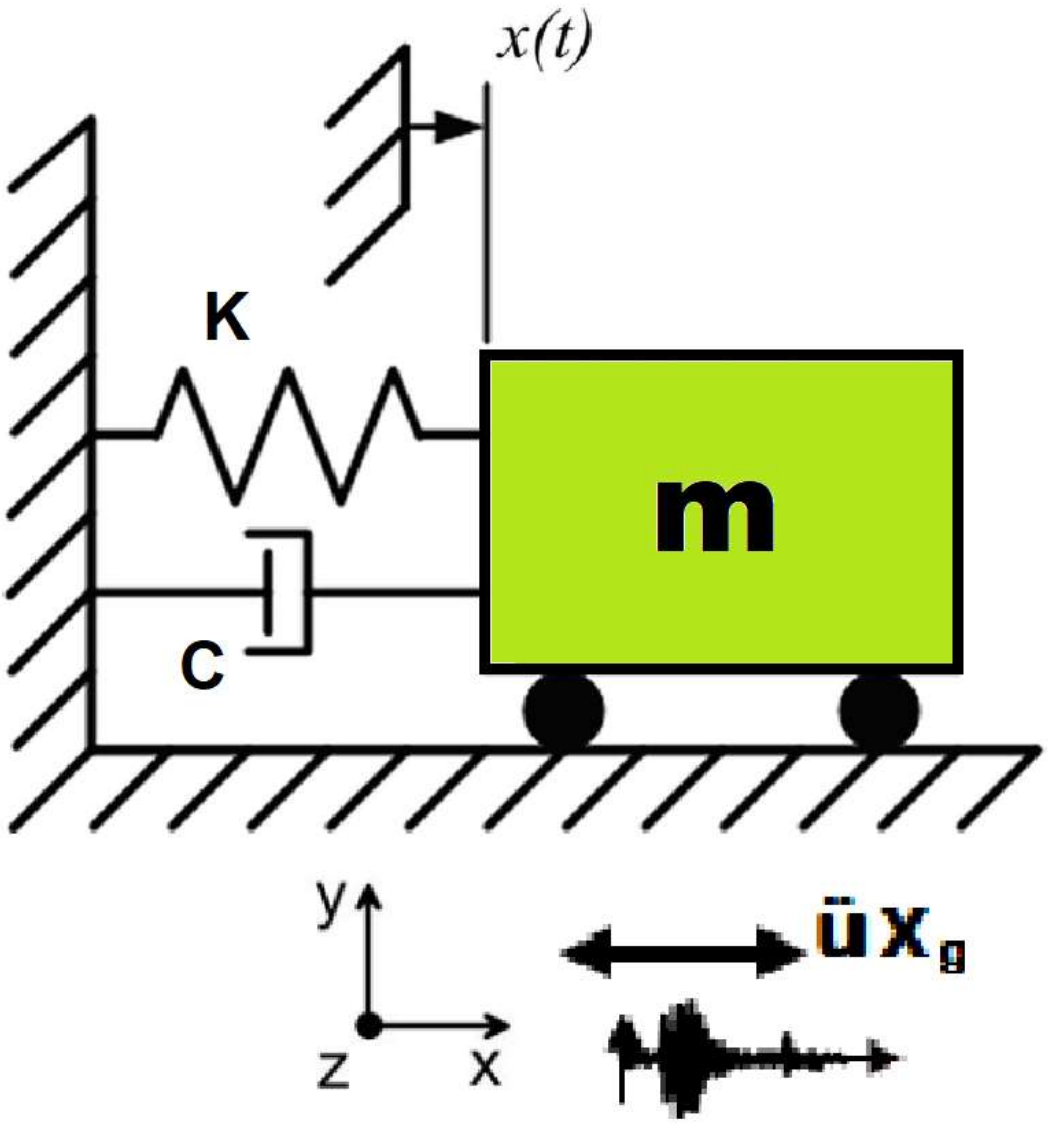
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

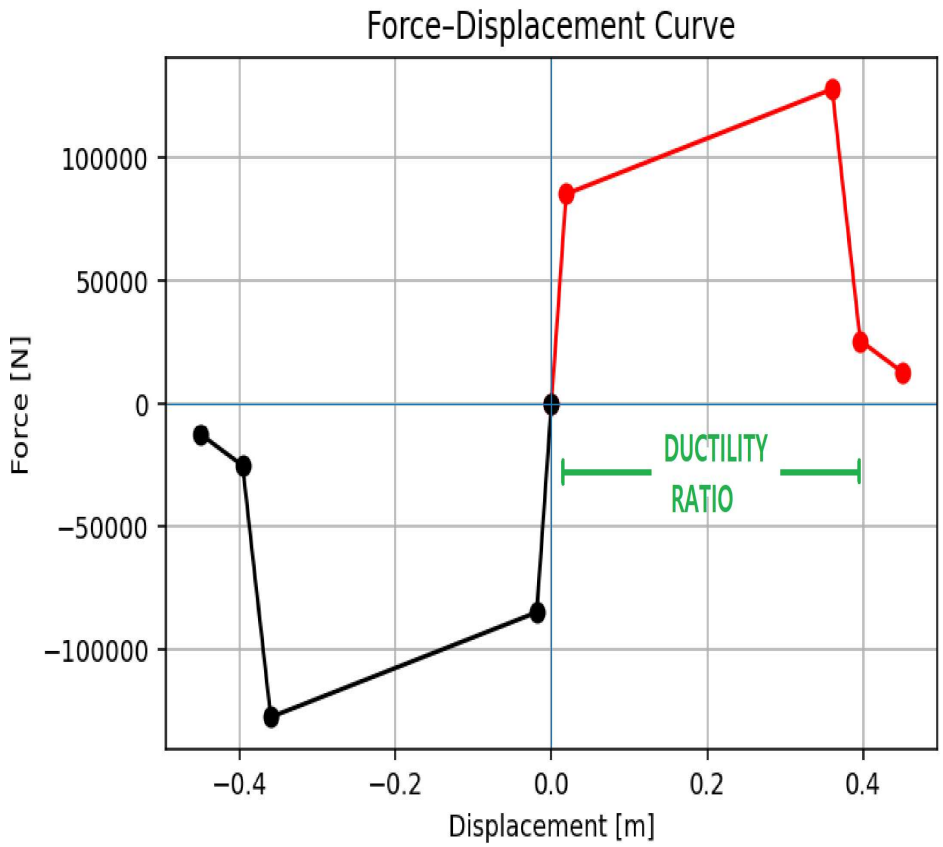
PARAMETRIC NONLINEAR SEISMIC DEMAND SPECTRA OF A SINGLE-DEGREE-OF-FREEDOM (SDOF) SYSTEMS WITH VARIABLE DUCTILITY CAPACITY IN OPENSEES

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

This OpenSees script performs a nonlinear SDOF ductility response-spectrum analysis that characterizes an oscillator's hysteretic behavior, energy dissipation, and damage under earthquake excitation. It defines mass, stiffness, yield strength, ultimate displacement, and damping; applies a cyclic displacement protocol to establish baseline energy dissipation; then runs nonlinear time-history analyses with Newmark integration over a parametric sweep of periods. For each period, it extracts peak displacement, velocity, acceleration, and base reaction, and computes the Ductility Damage Index, Energy Dissipation Capacity Index, and Equivalent Viscous Damping Ratio. Finally, it statistically processes and plots displacement, velocity, acceleration, base-reaction, DDI, EDCI, and EVDR spectra, generates fragility information, and exports results for performance-based earthquake engineering and parametric studies.

Since the structure was assumed to be inelastic and the response spectra were then plotted for the energy dissipation capacity index and the ductility damage index, this shows the natural period beyond which an increase in the structure's period leads to a higher probability of damage, and as the ductility ratio of the structure increases, the level of damage decreases.


x(t)
K
m
C
y
z
x
ü x g


Force-Displacement Curve
Force [N]
Displacement [m]
100000
50000
0
−50000
−100000
−0.4
−0.2
0.0
0.2
0.4
DUCTILITY
RATIO

$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CI_EVDR.+\SDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR.py

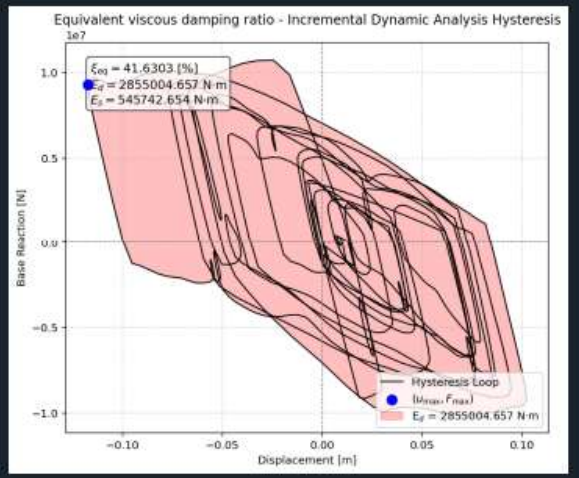
SDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR.py | EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN.py

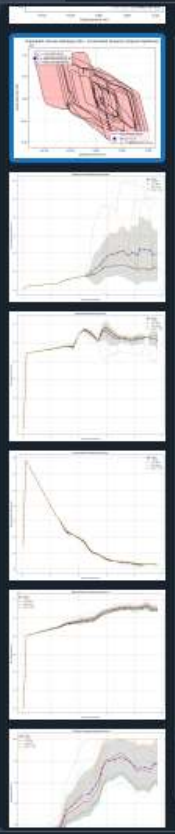
```
#########################################################################################################
#                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#     NONLINEAR DYNAMIC ANALYSIS OF A SINGLE-DEGREE-OF-FREEDOM (MDOF) SOIL STRUCTURE SYSTEM UTILIZING
#          GROUND MOTIONS IN OPENSEES AND EVALUATION OF CONSTANT DUCTILITY SEISMIC RESPONSE SPECTRUM
#---------------------------------------------------------------------------------------------------------
#                                    INELASTIC RESPONSE SPECTRUM ANALYSIS
#---------------------------------------------------------------------------------------------------------
#                EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
#---------------------------------------------------------------------------------------------------------
#          ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
#---------------------------------------------------------------------------------------------------------
# This program performs  Dynamic Analysis on a Single-Degree-of-Freedom (SDOF) system
# subjected to 1 seismic ground motions. The analysis evaluates the structural response under varying
# levels of seismic intensity.
# The framework is designed to support researchers and engineers in assessing the probabilistic seismic
# performance of structures, with a focus on understanding the impact of uncertainty on structural
# response and design.
#---------------------------------------------------------------------------------------------------------
# Key Features:
# - Simulation of SDOF system using OpenSees.
# - Scaling of ground motions for Nonlinear Dynamic Analysis.
# - Probabilistic fragility assessment based on predefined damage states.
# - Visualization of structural response and fragility curves.
# - Export of results for further analysis.
#---------------------------------------------------------------------------------------------------------
#                         THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                    EMAIL: salar.d.ghashghaei@gmail.com
#########################################################################################################

"""
This code implements a comprehensive nonlinear dynamic analysis framework for
performance-based earthquake engineering assessment of single-degree-of-freedom
(SDOF) systems. The methodology combines traditional nonlinear time-history
analysis with modern probabilistic and machine learning techniques for advanced
```

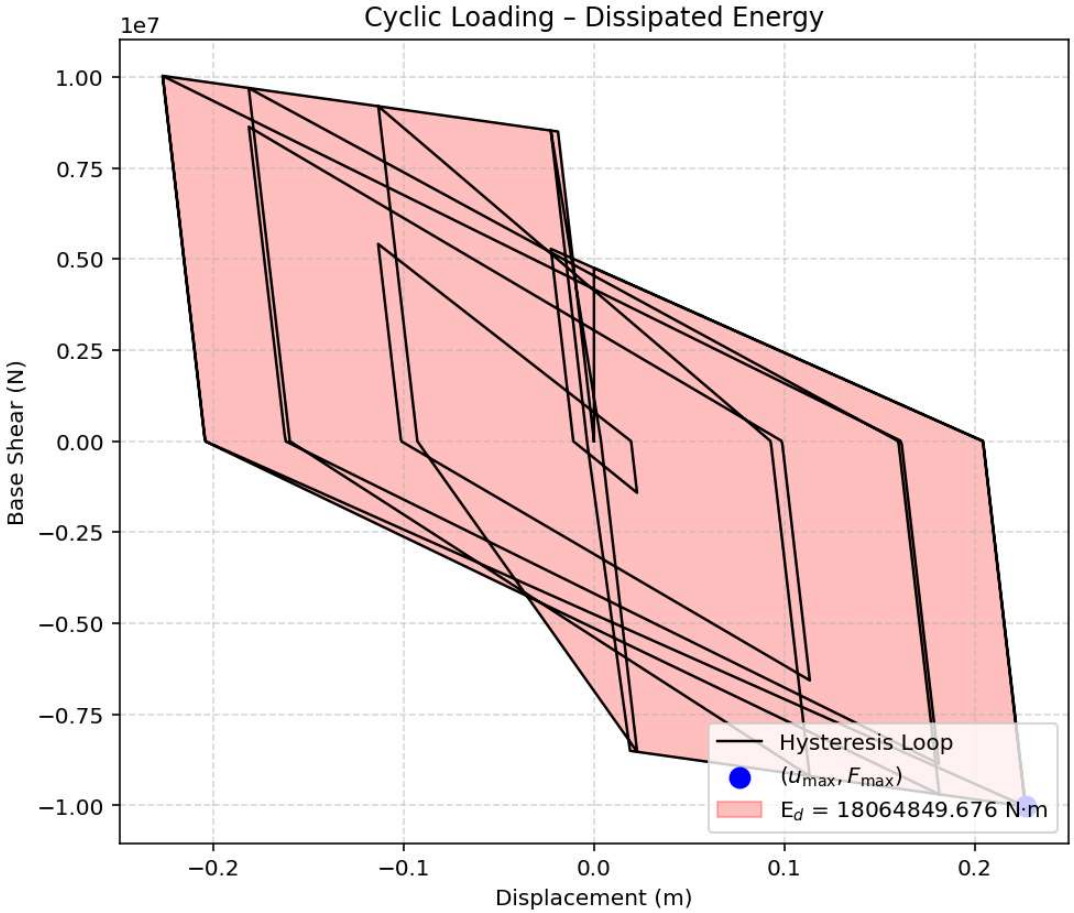
..DEX\11_SDOF_8_ANA\SDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR+

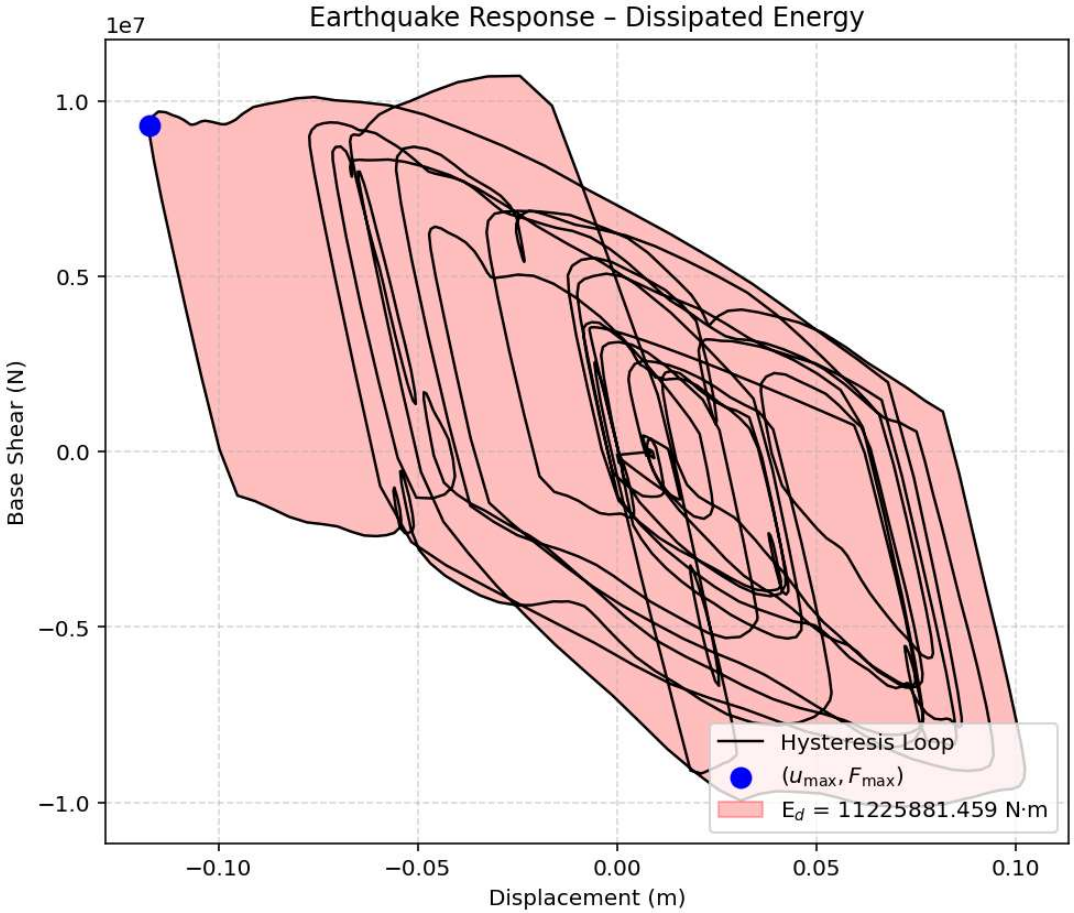
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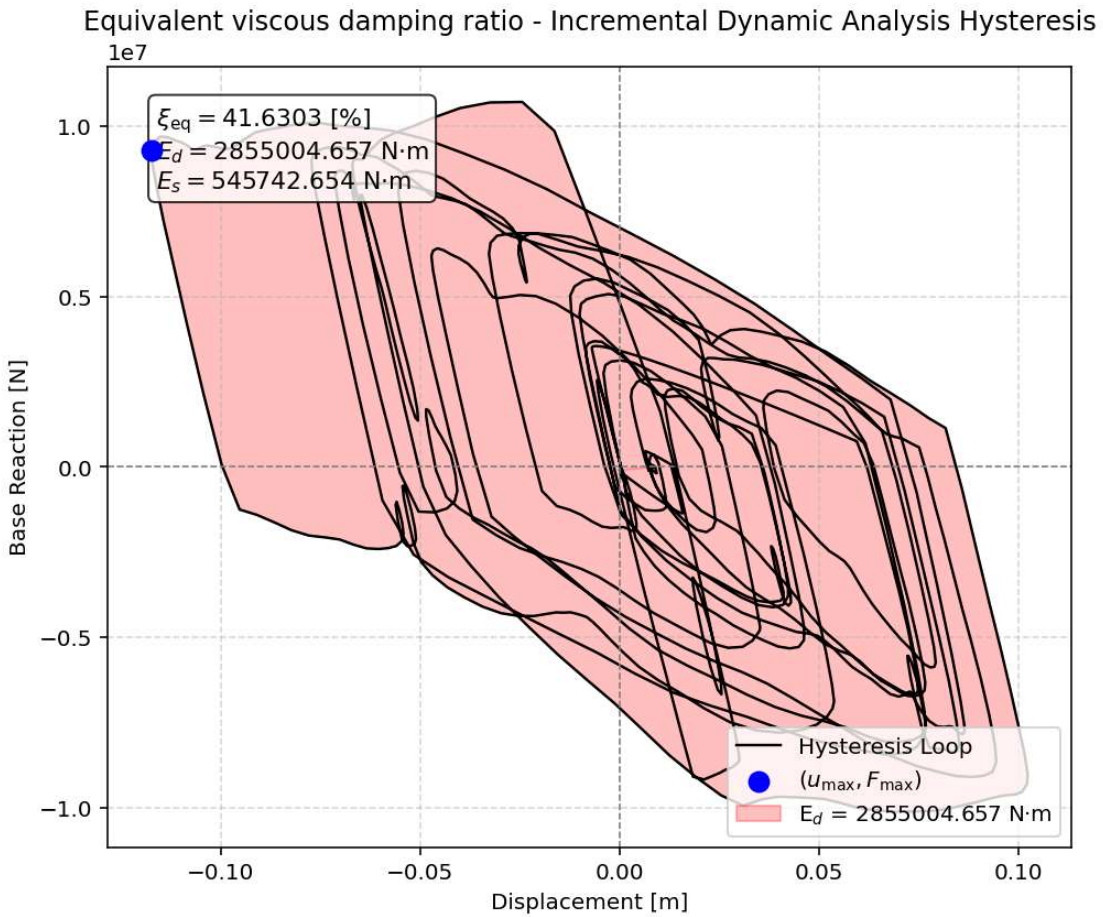


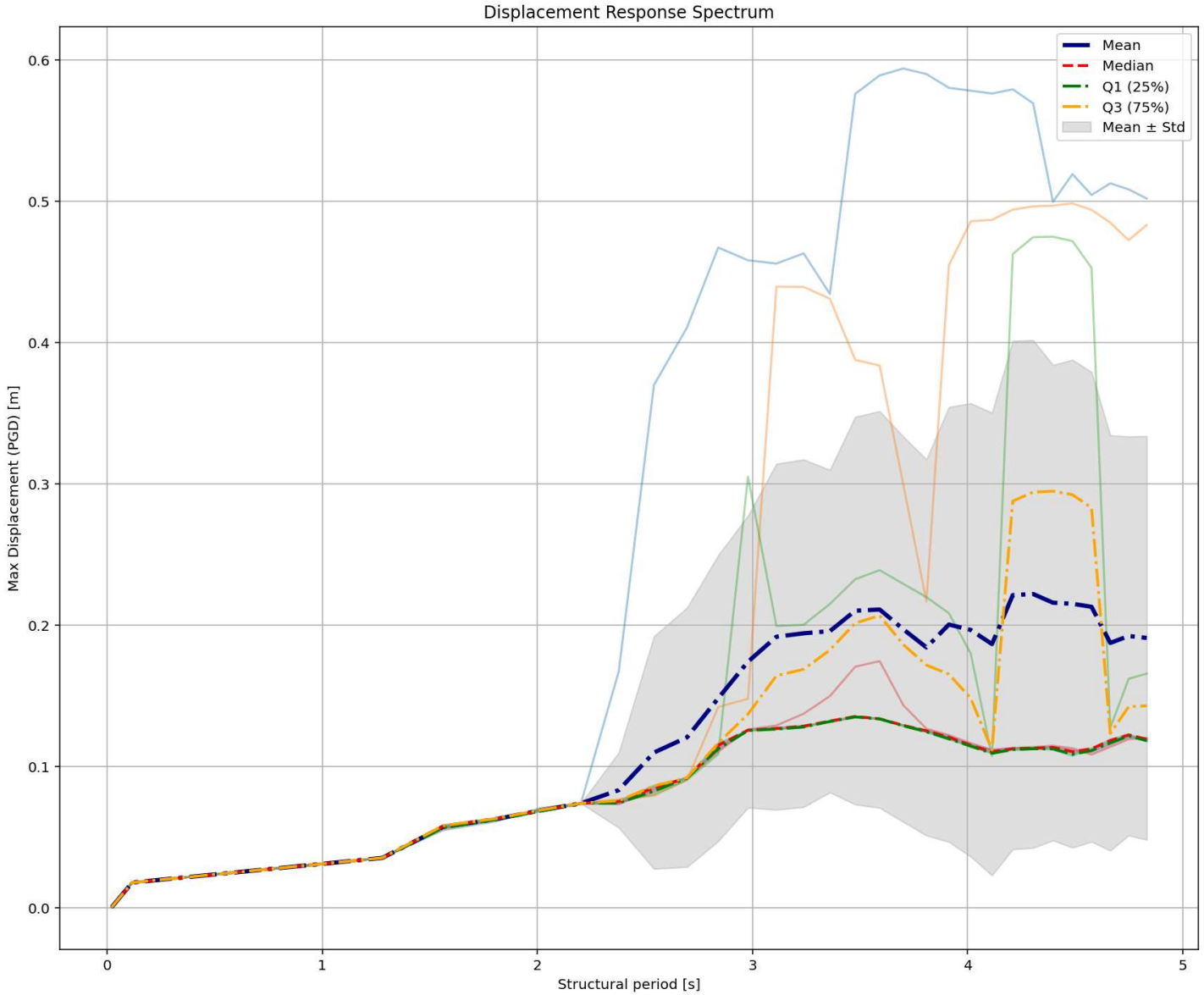


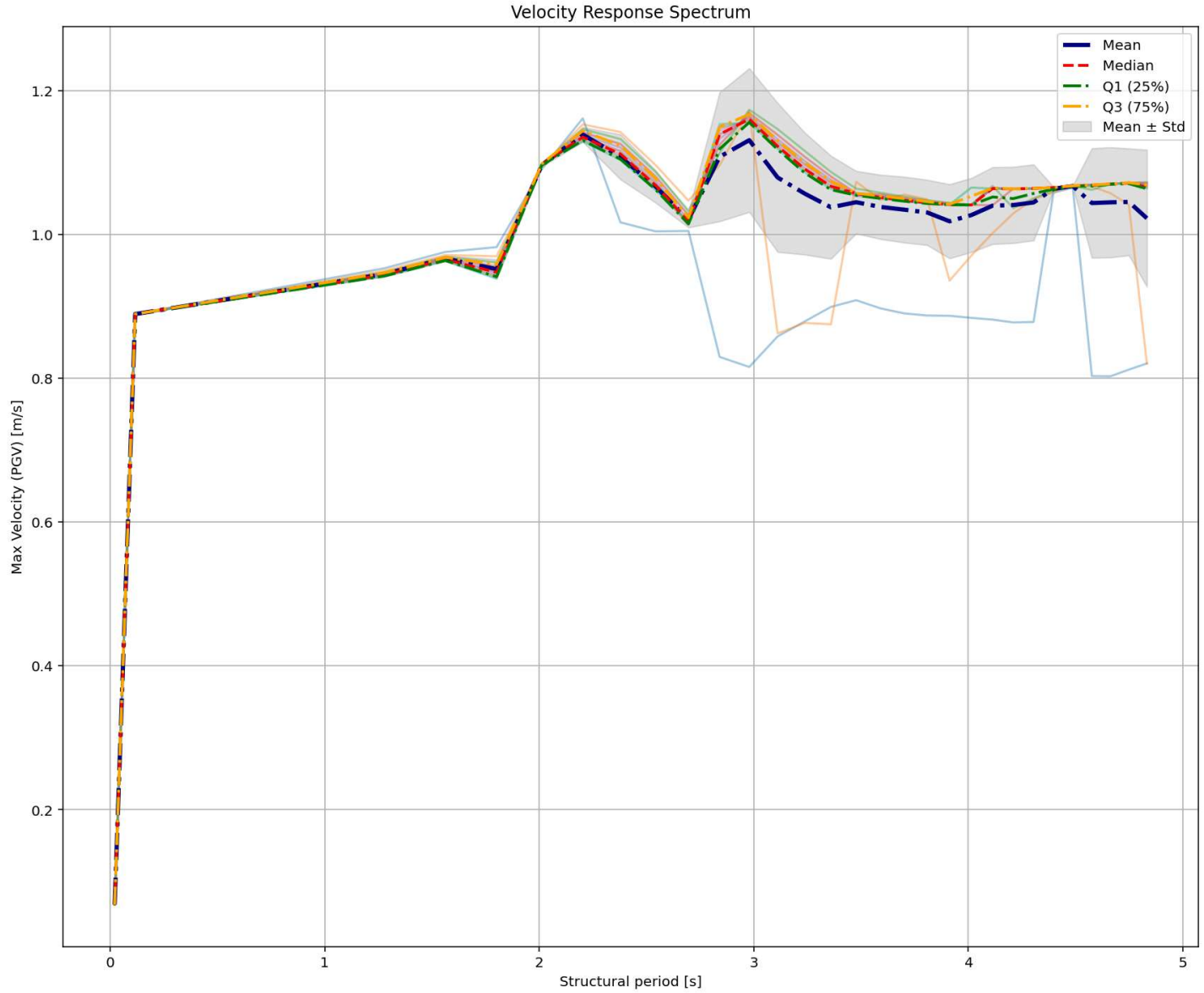
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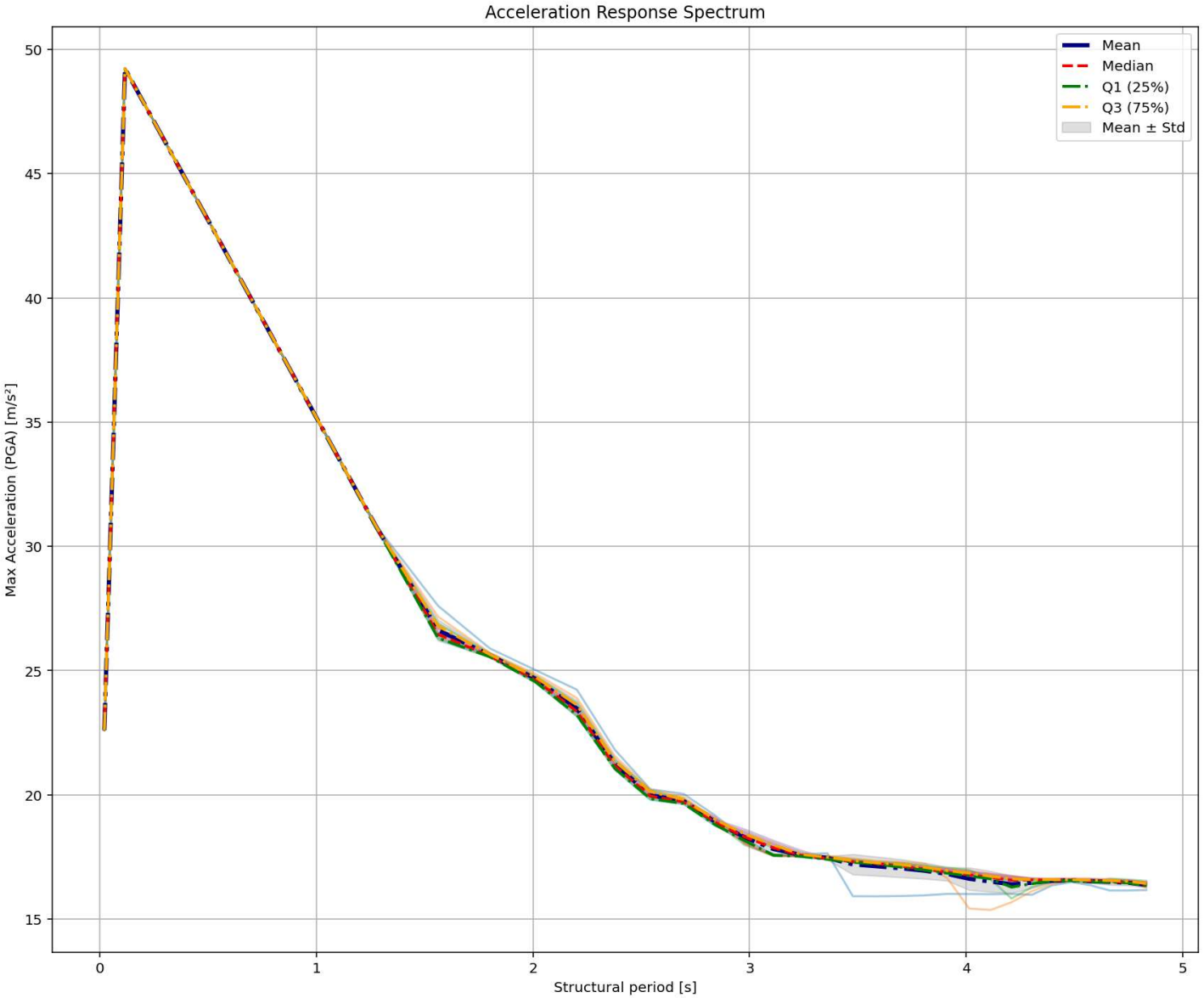

Cyclic Loading – Dissipated Energy
1e7
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Base Shear (N)
−0.2
−0.1
0.0
0.1
0.2
Displacement (m)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 18064849.676 N·m

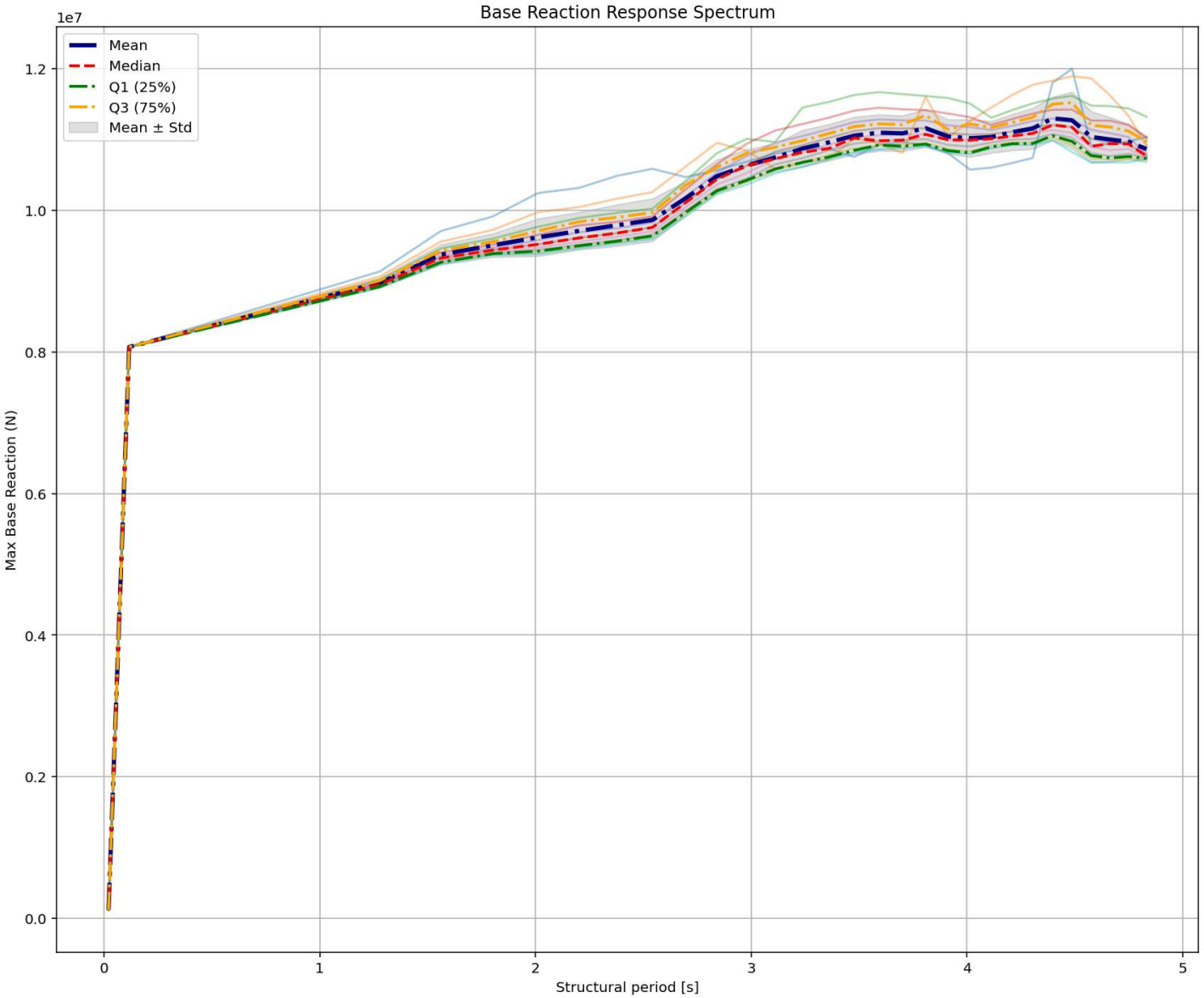

Earthquake Response – Dissipated Energy
1e7
1.0
0.5
0.0
−0.5
−1.0
Base Shear (N)
−0.10
−0.05
0.00
0.05
0.10
Displacement (m)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 11225881.459 N·m

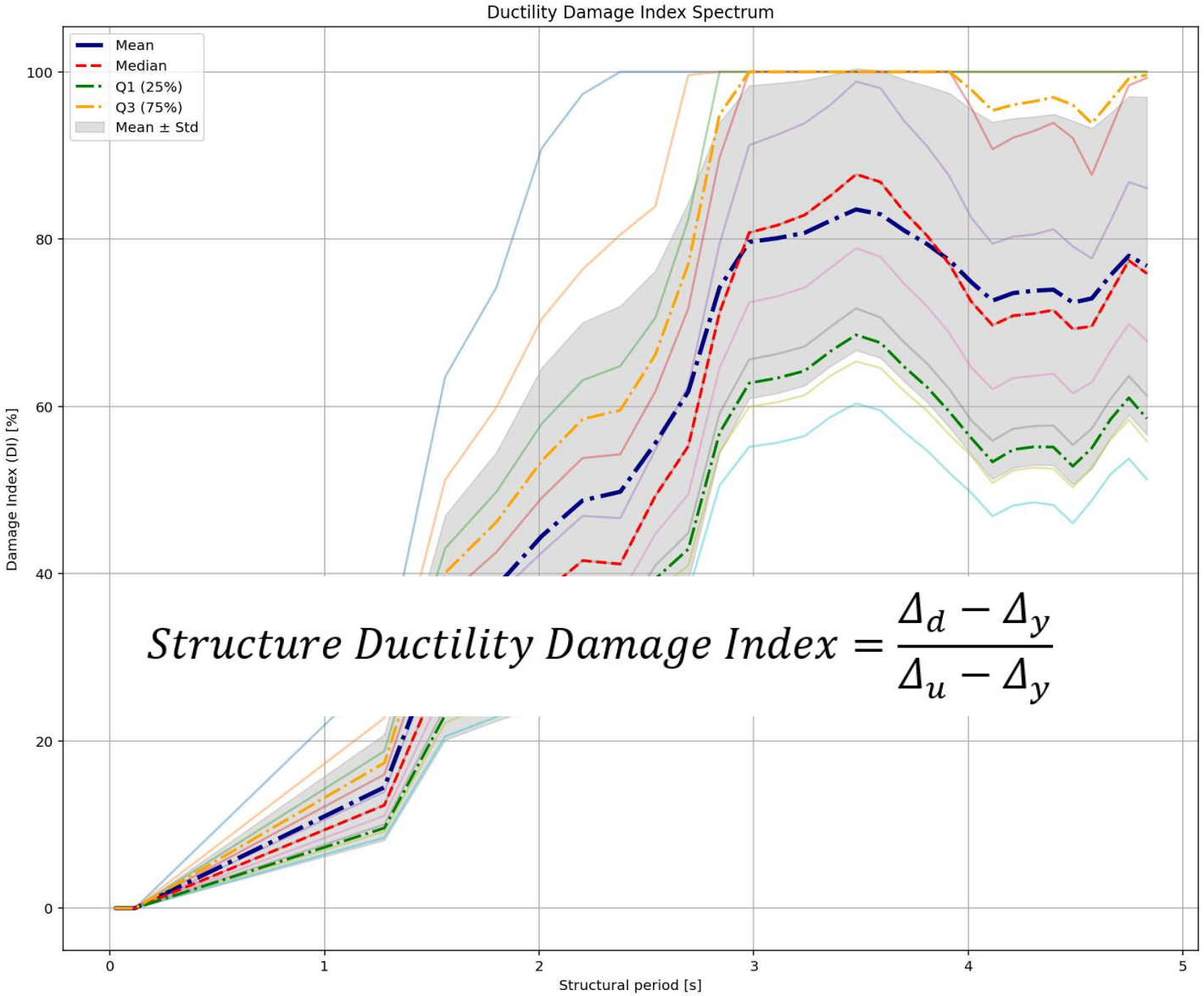

Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis
1e7
$\xi_{eq} = 41.6303$ [%]
$E_d = 2855004.657$ N·m
$E_s = 545742.654$ N·m
Base Reaction [N]
Displacement [m]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 2855004.657 N·m


Displacement Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Displacement (PGD) [m]
Structural period [s]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
1
2
3
4
5

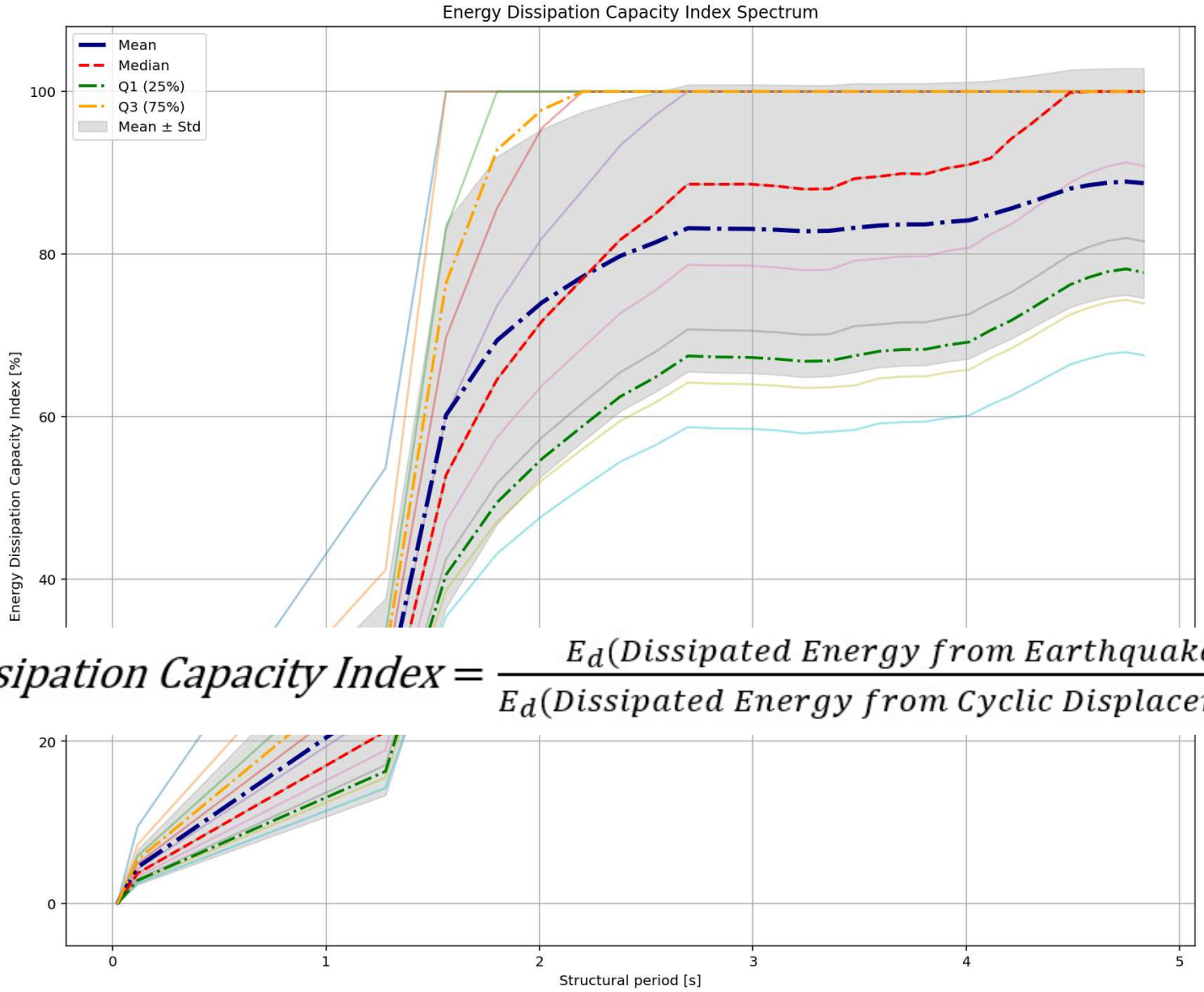

Velocity Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Velocity (PGV) [m/s]
Structural period [s]
0.2
0.4
0.6
0.8
1.0
1.2
0
1
2
3
4
5


Acceleration Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Acceleration (PGA) [m/s²]
Structural period [s]
50
45
40
35
30
25
20
15
0
1
2
3
4
5

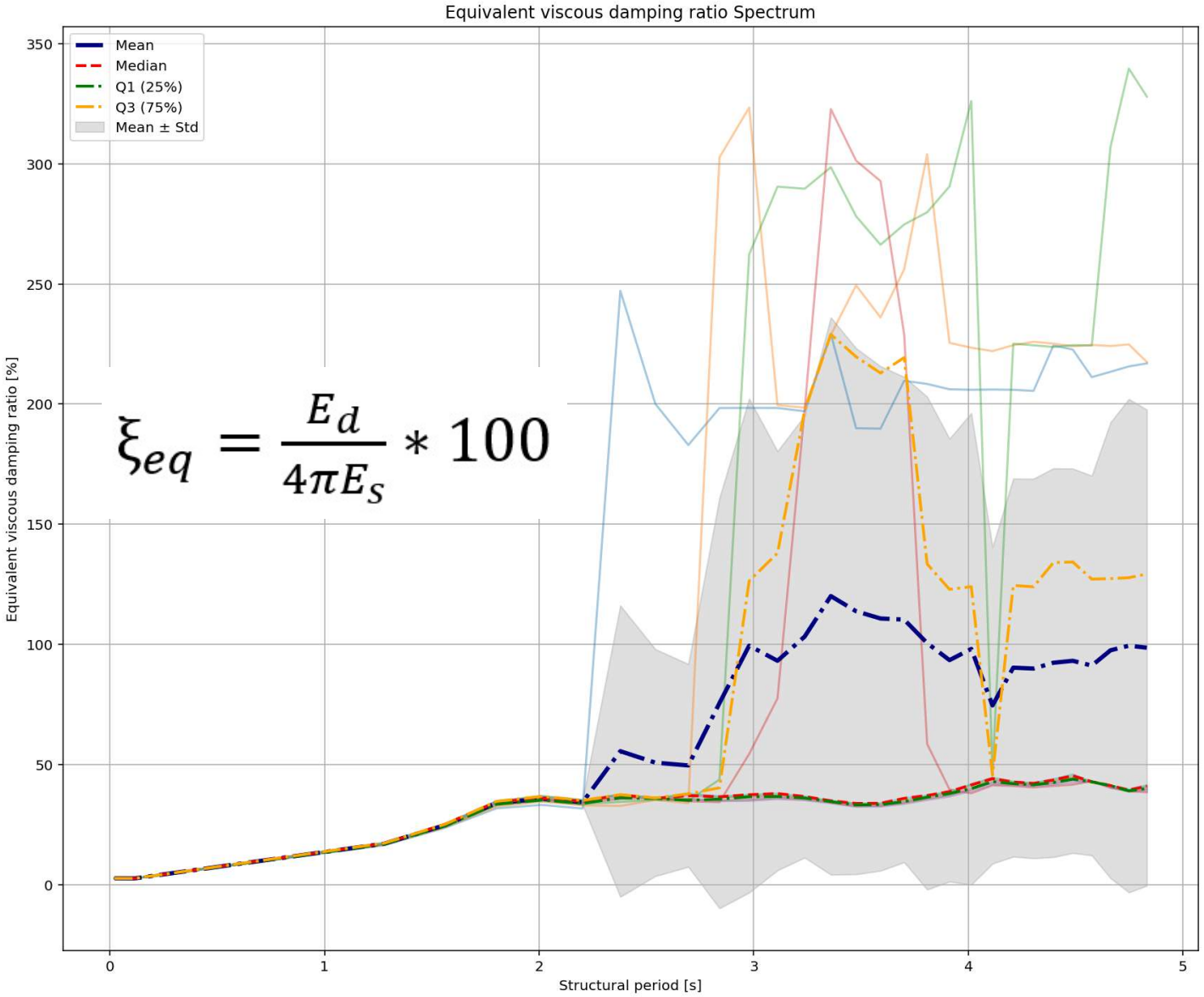

Base Reaction Response Spectrum
1e7
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Base Reaction (N)
Structural period [s]
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0
1
2
3
4
5

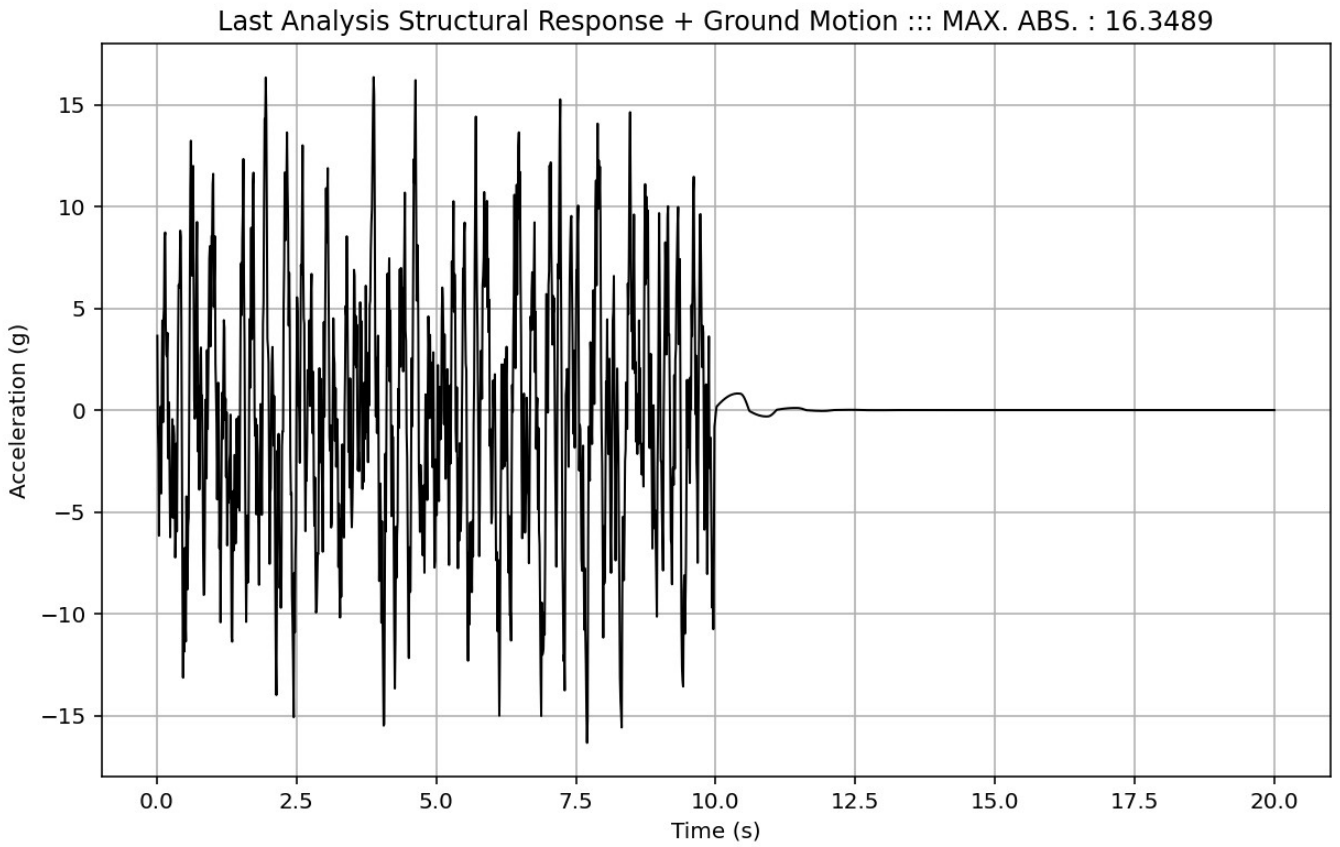


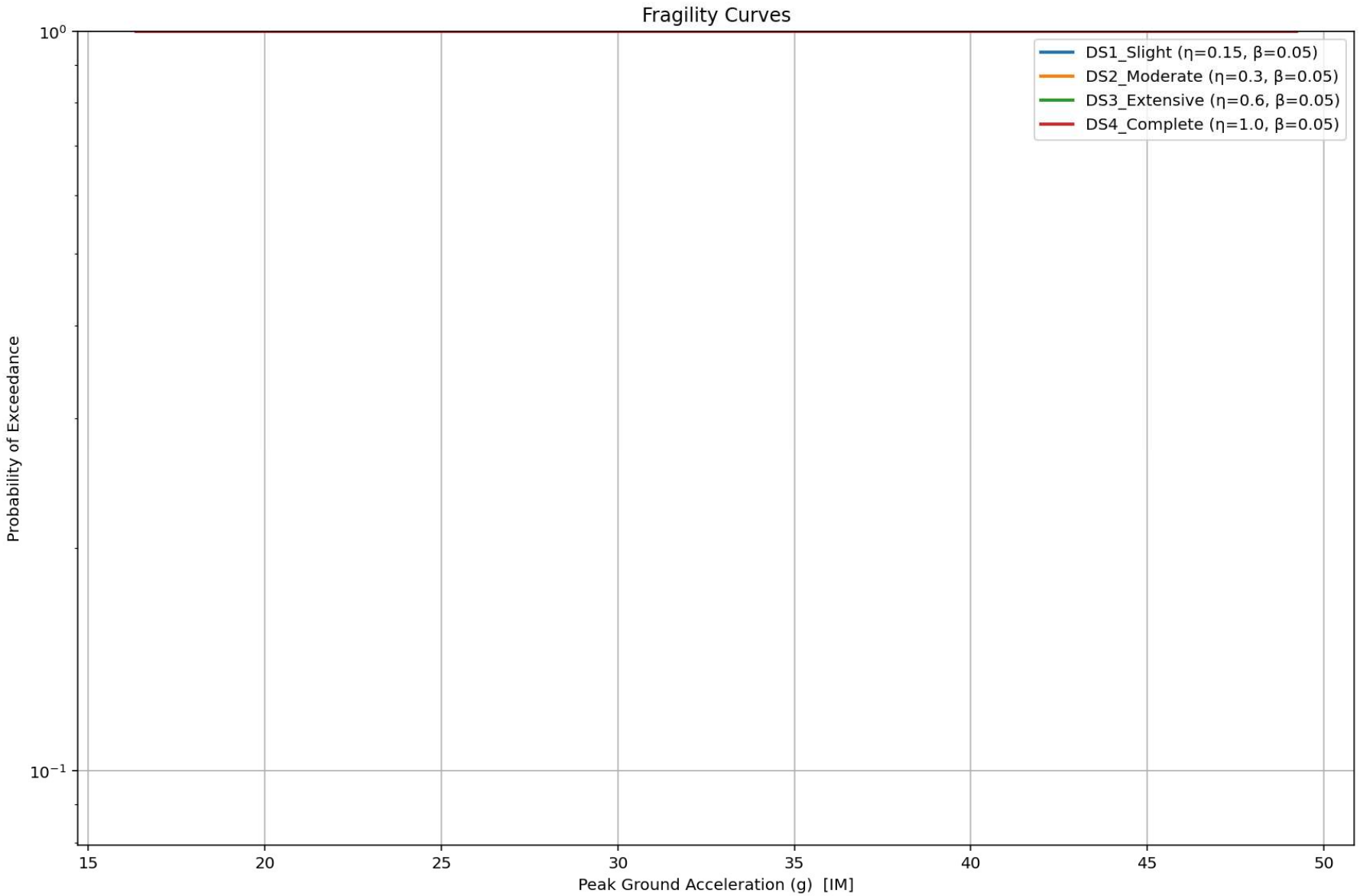
$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

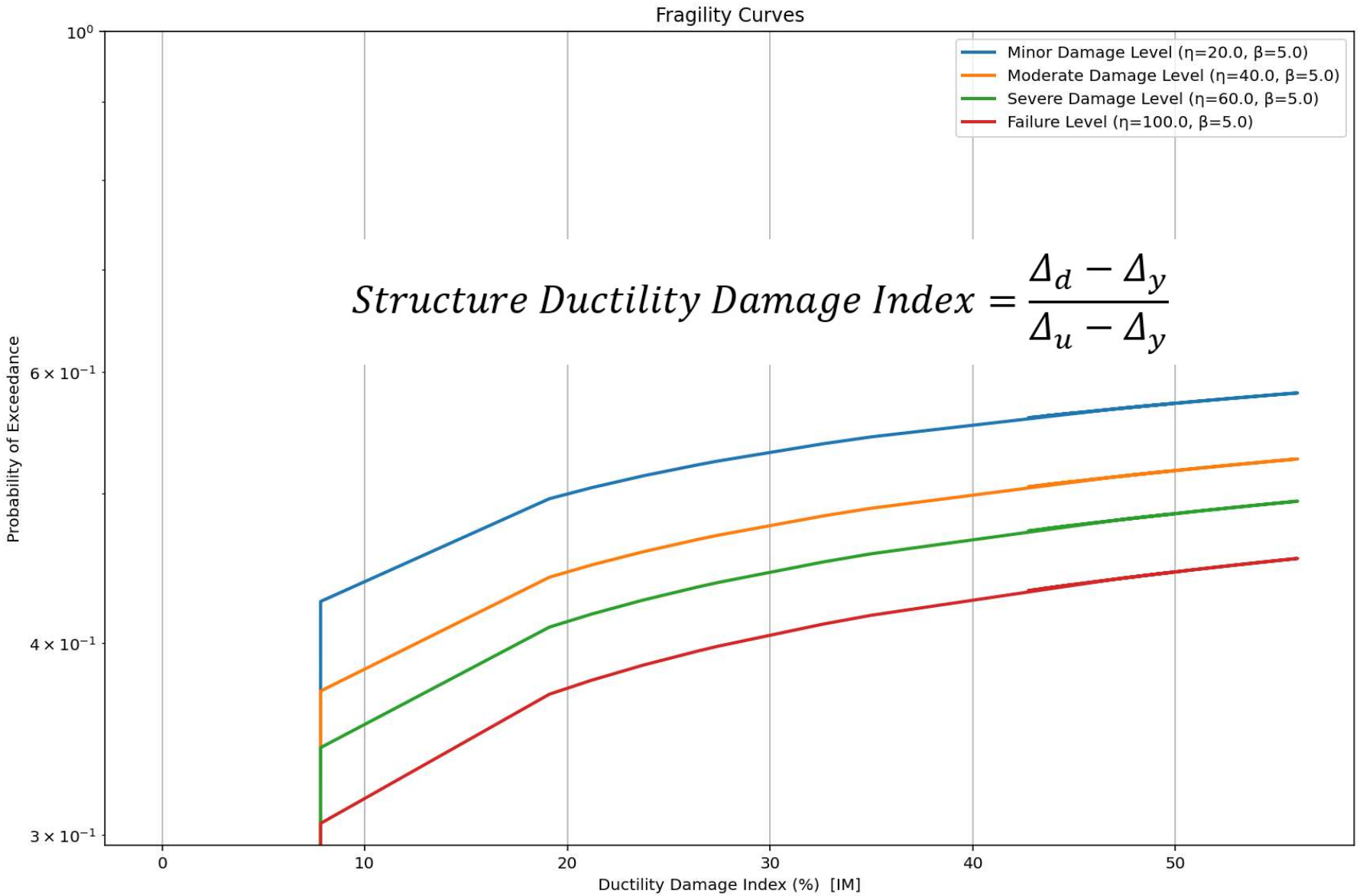


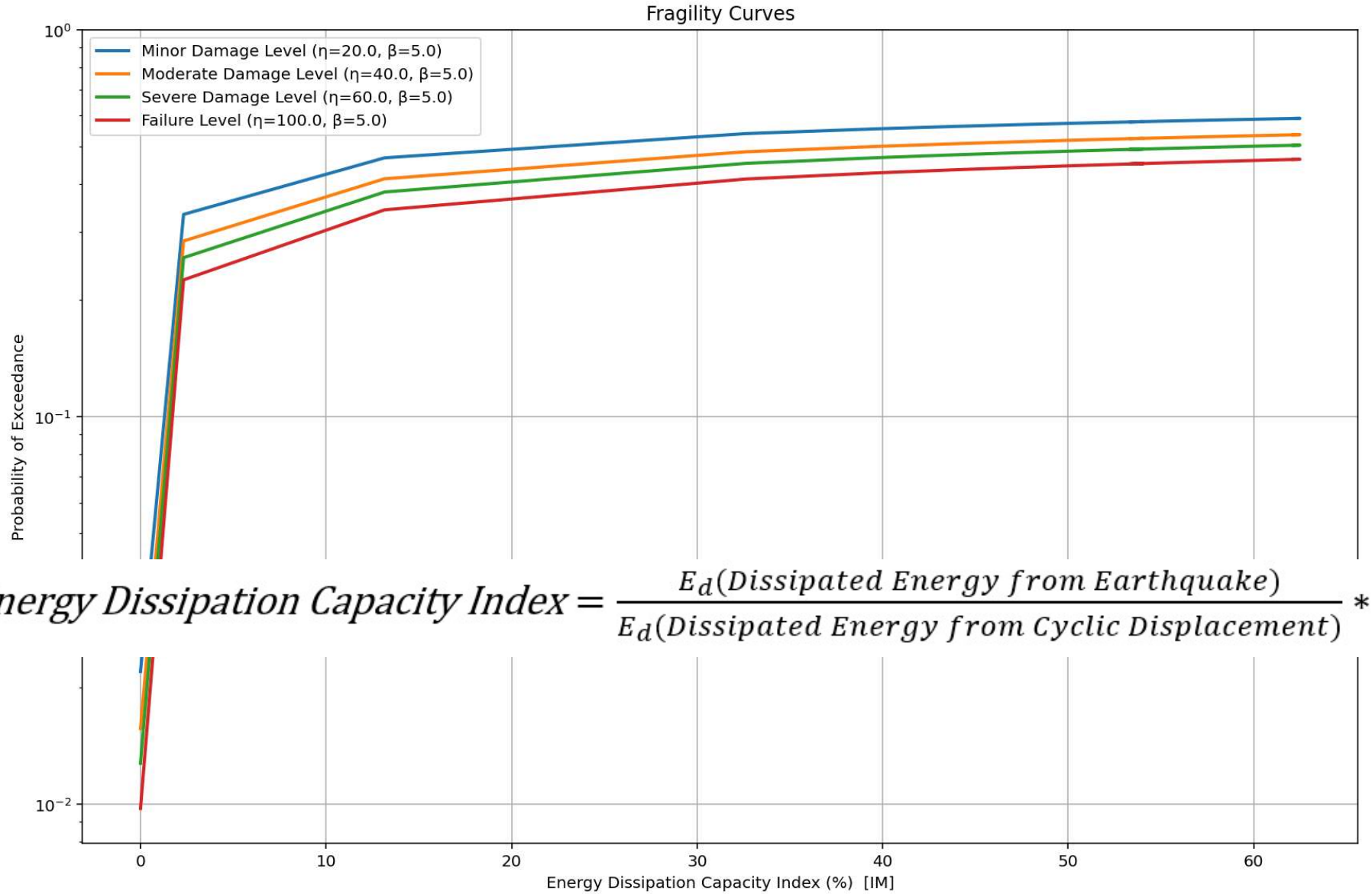
$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 100$$


Equivalent viscous damping ratio Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$
Equivalent viscous damping ratio [%]
Structural period [s]
0
50
100
150
200
250
300
350
0
1
2
3
4
5

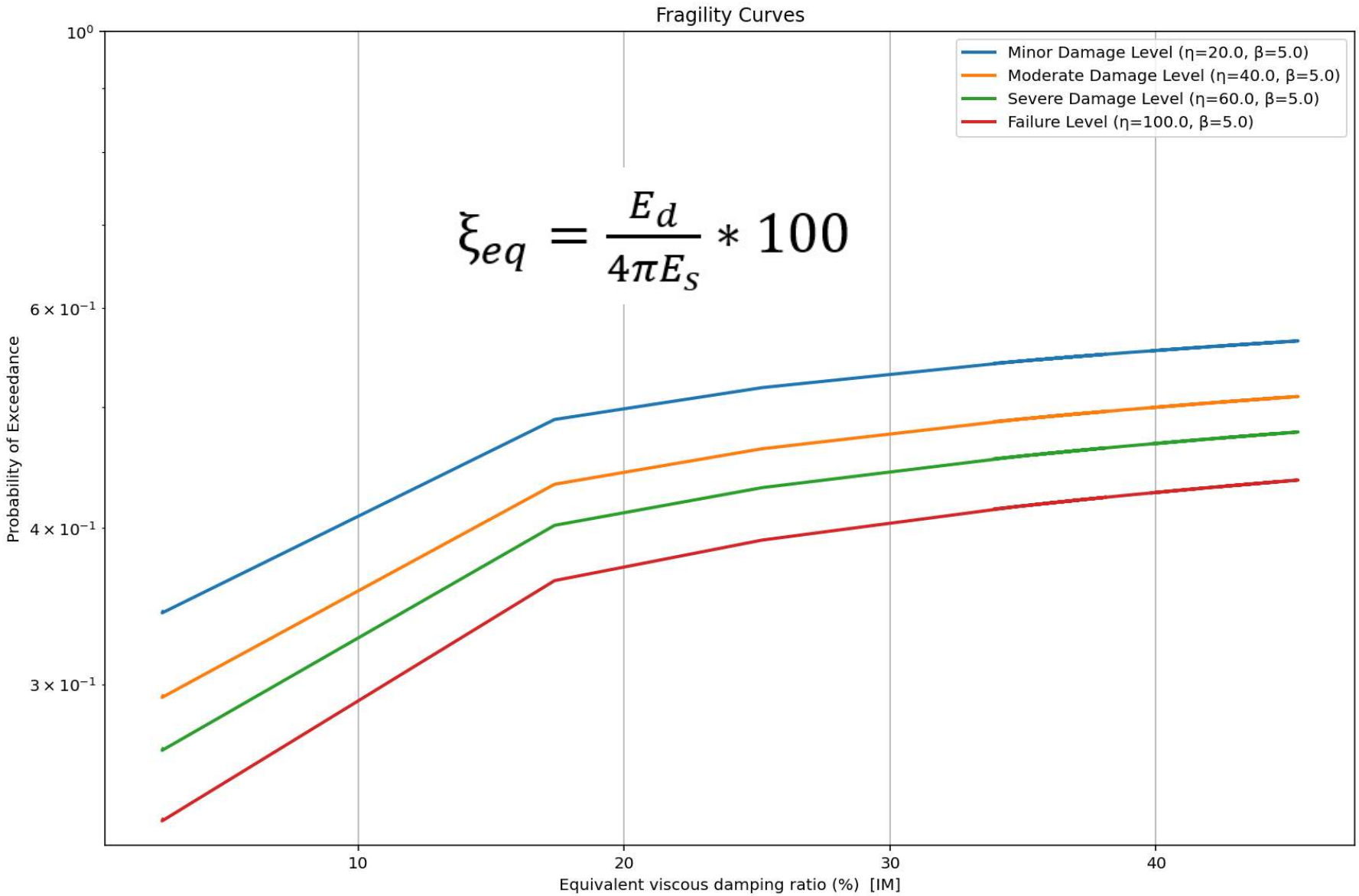

Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 16.3489
Acceleration (g)
15
10
5
0
−5
−10
−15
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

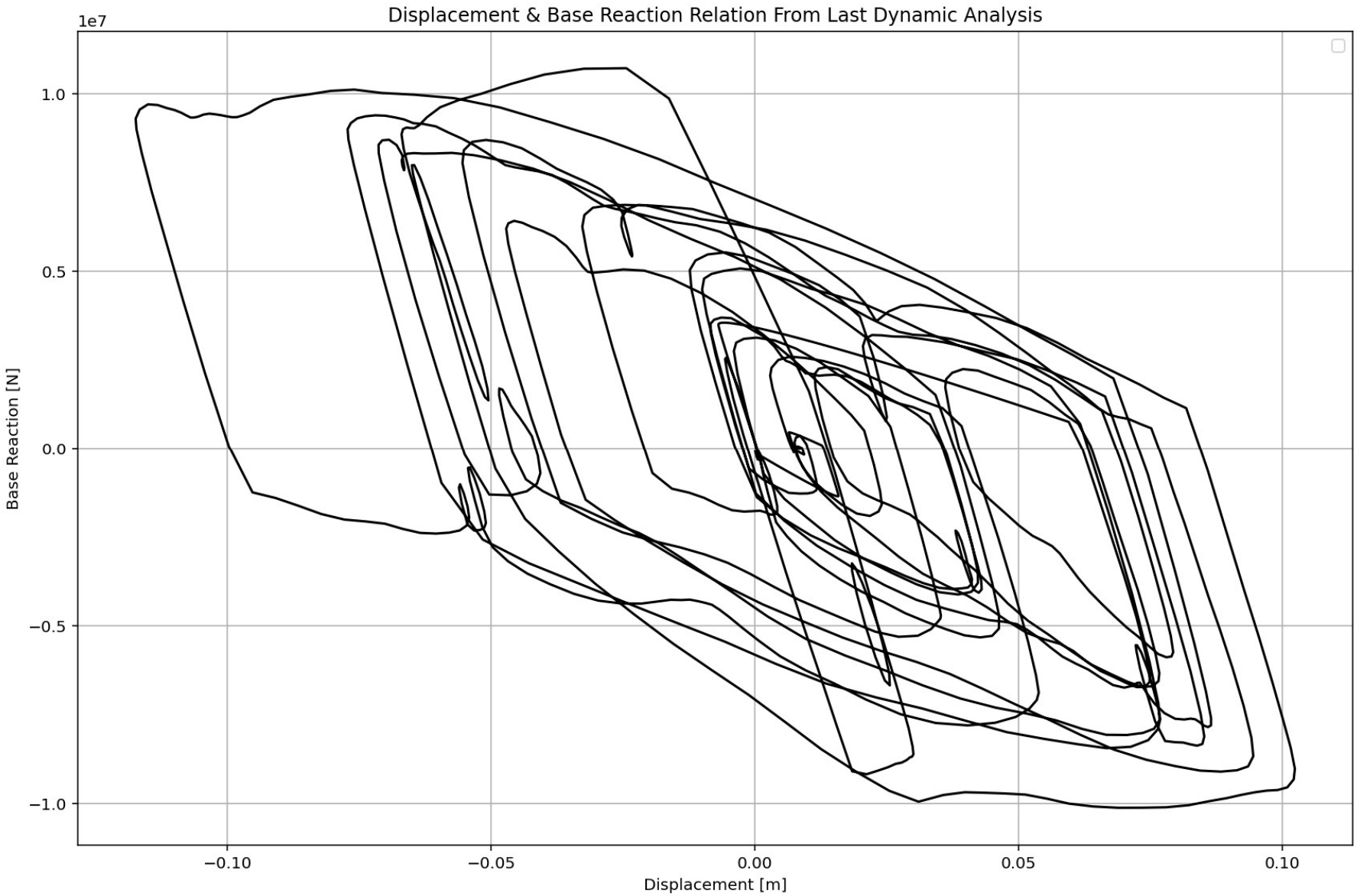

Fragility Curves
DS1_Slight (η=0.15, β=0.05)
DS2_Moderate (η=0.3, β=0.05)
DS3_Extensive (η=0.6, β=0.05)
DS4_Complete (η=1.0, β=0.05)
Probability of Exceedance
10^0
10^{-1}
15
20
25
30
35
40
45
50
Peak Ground Acceleration (g) [IM]

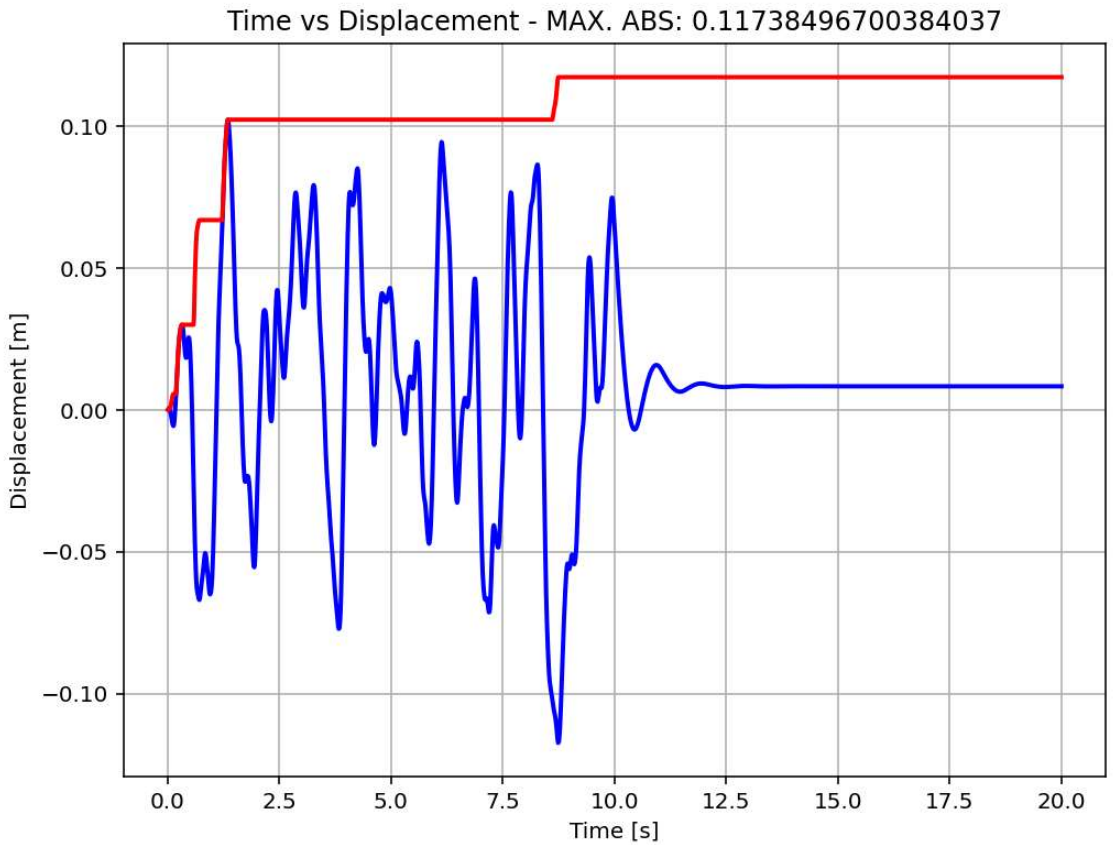

Fragility Curves
Minor Damage Level (η=20.0, β=5.0)
Moderate Damage Level (η=40.0, β=5.0)
Severe Damage Level (η=60.0, β=5.0)
Failure Level (η=100.0, β=5.0)
$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$
Probability of Exceedance
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
0
10
20
30
40
50
Ductility Damage Index (%) [IM]

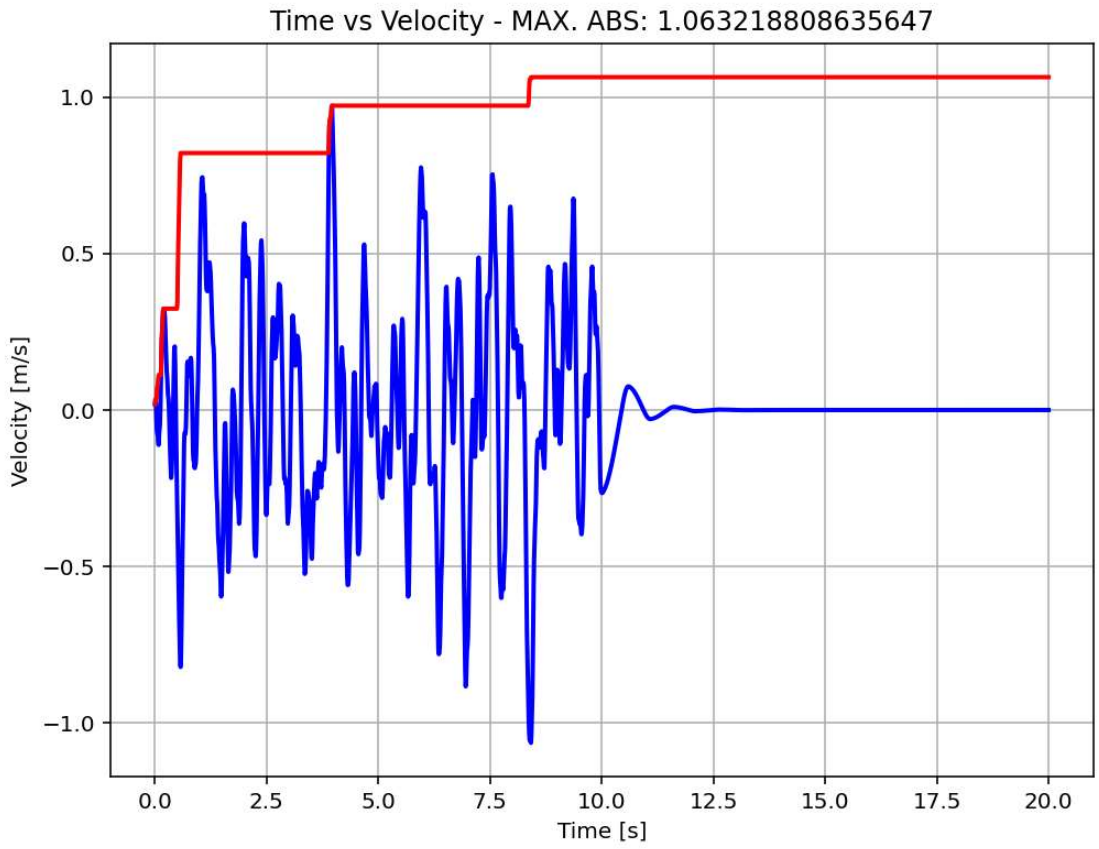


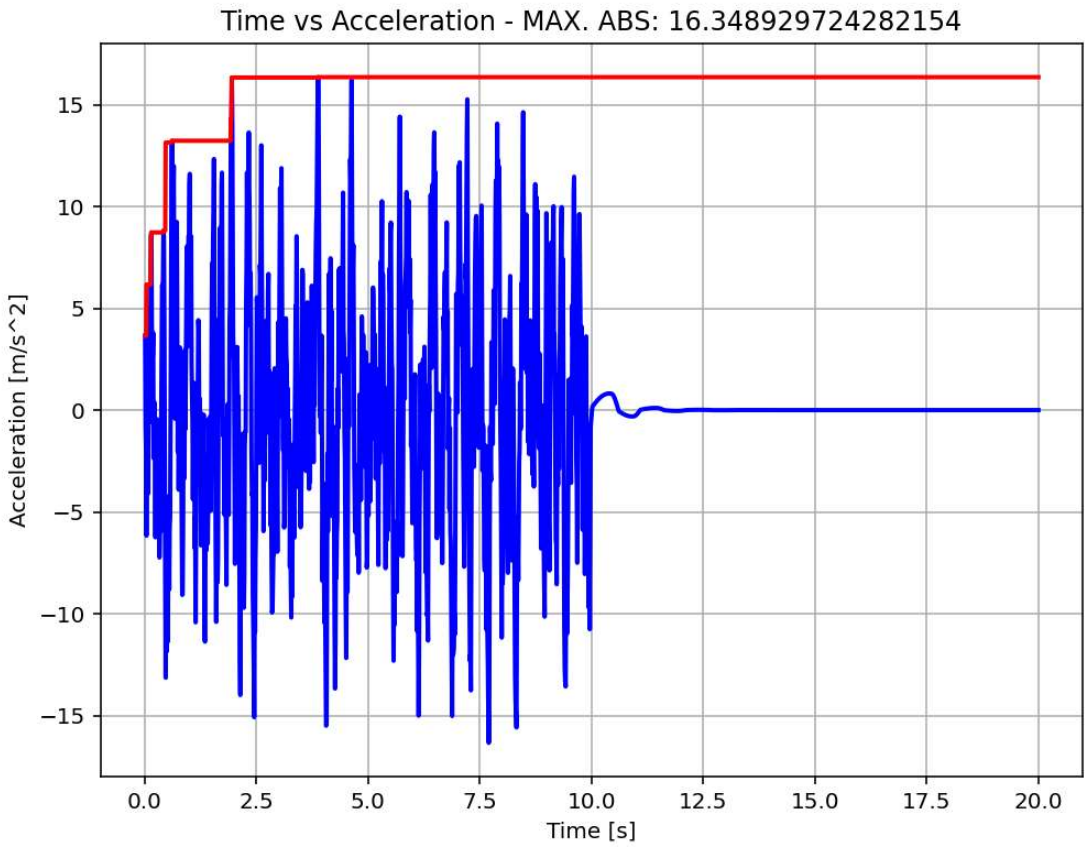
$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 100$$

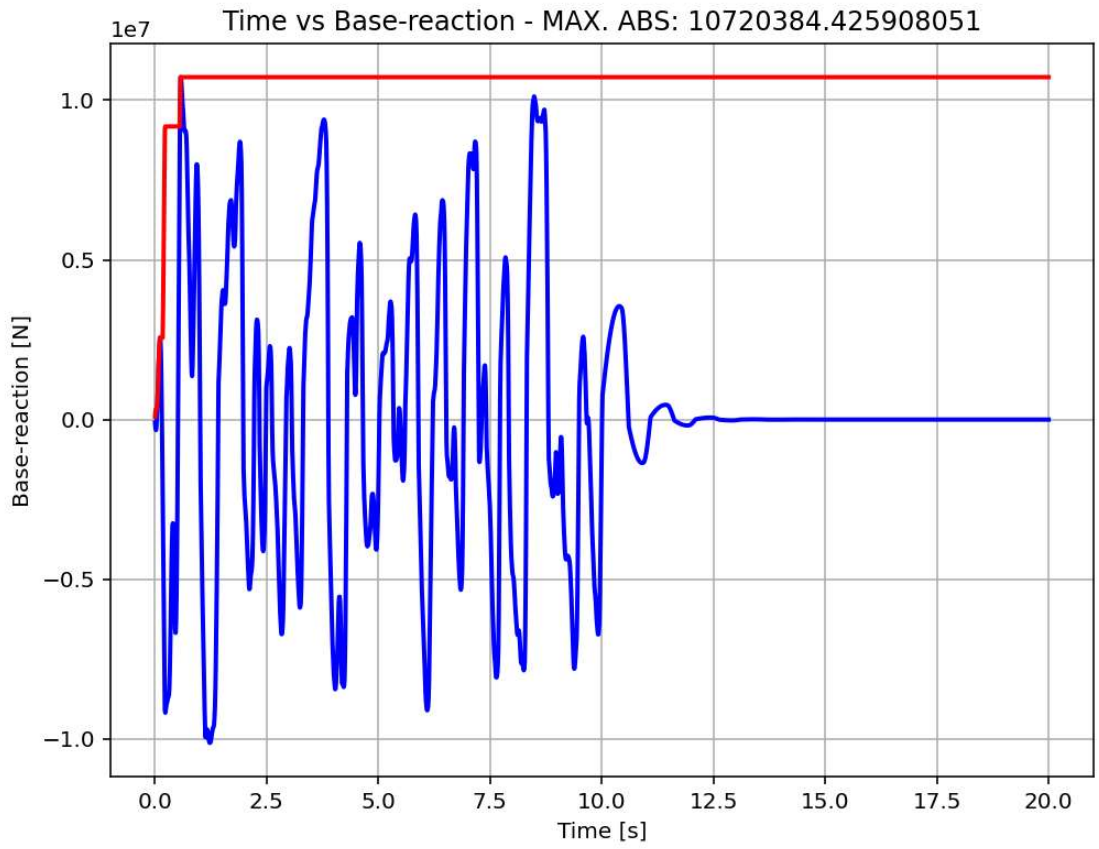

Fragility Curves
$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$
Minor Damage Level (η=20.0, β=5.0)
Moderate Damage Level (η=40.0, β=5.0)
Severe Damage Level (η=60.0, β=5.0)
Failure Level (η=100.0, β=5.0)
Probability of Exceedance
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
10
20
30
40
Equivalent viscous damping ratio (%) [IM]

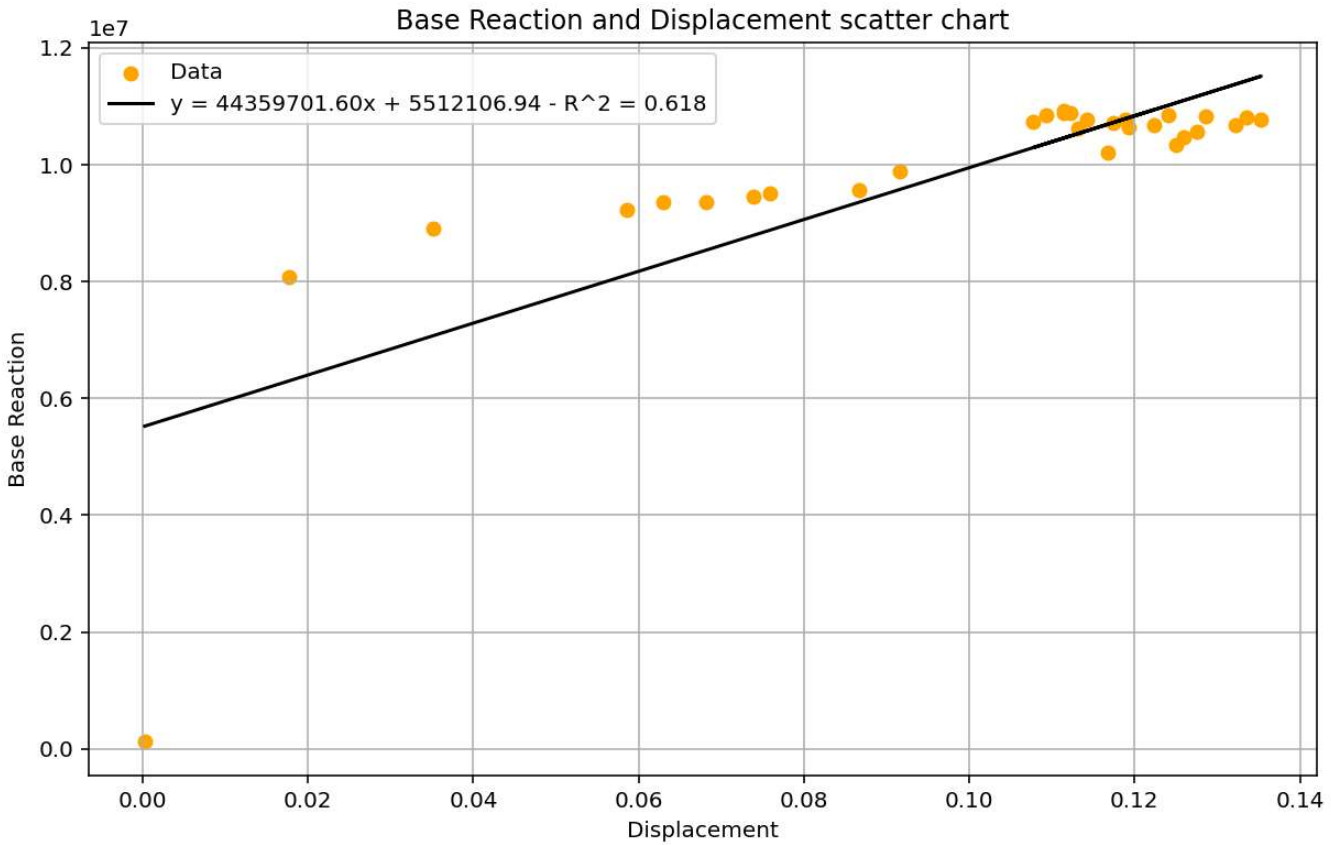

Displacement & Base Reaction Relation From Last Dynamic Analysis
1e7
1.0
0.5
0.0
−0.5
−1.0
Base Reaction [N]
−0.10
−0.05
0.00
0.05
0.10
Displacement [m]

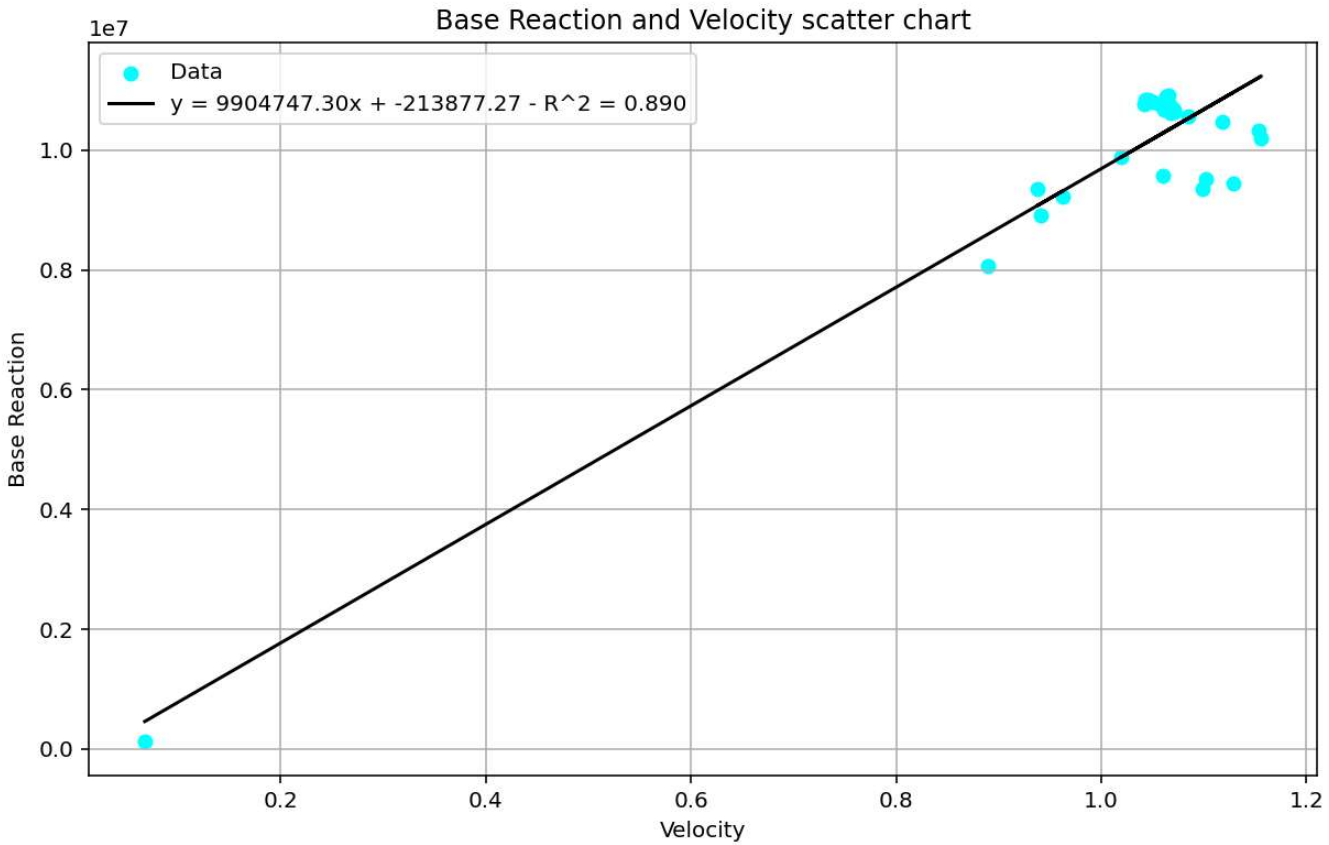

Time vs Displacement - MAX. ABS: 0.11738496700384037
Displacement [m]
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

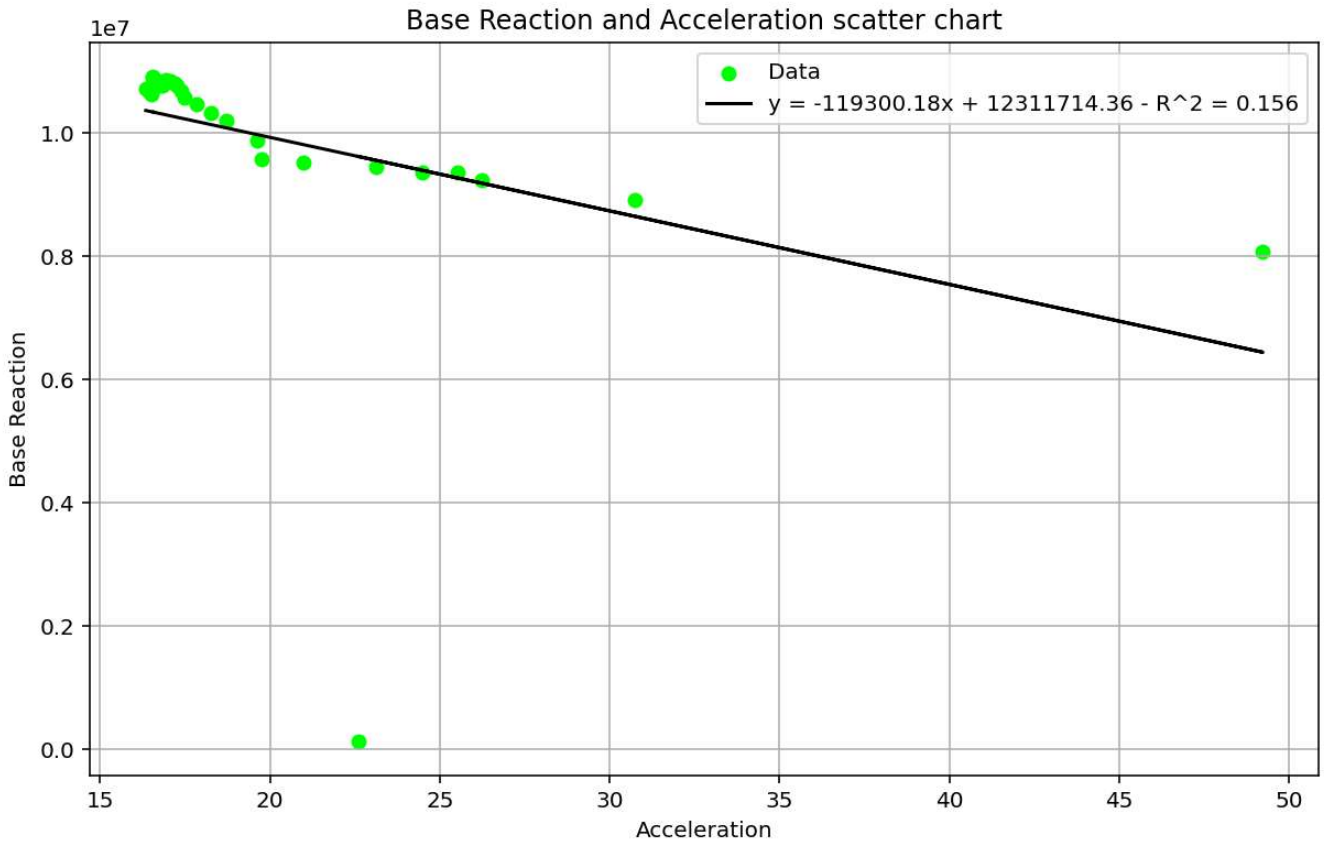

Time vs Velocity - MAX. ABS: 1.063218808635647
Velocity [m/s]
Time [s]
1.0
0.5
0.0
−0.5
−1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

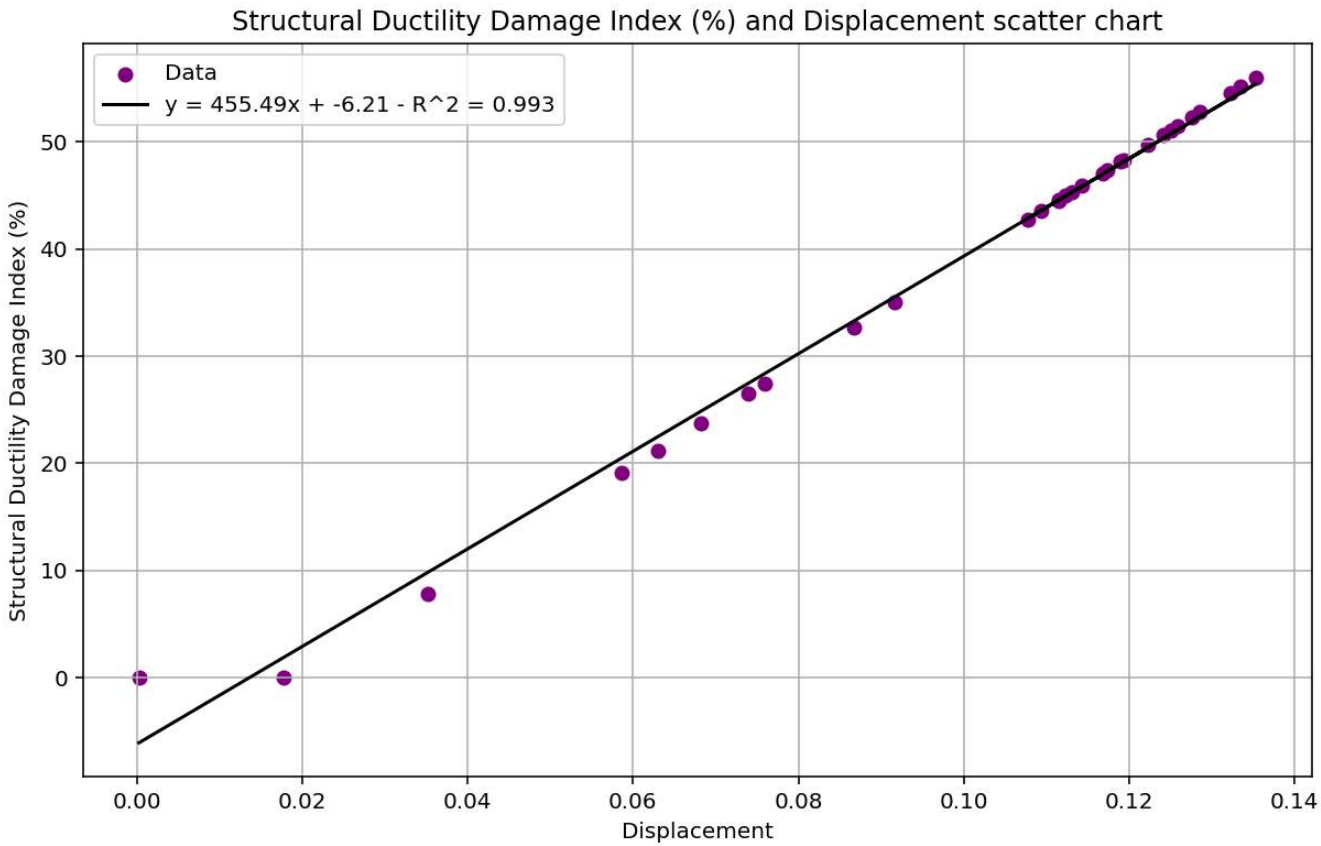

Time vs Acceleration - MAX. ABS: 16.348929724282154
Acceleration [m/s^2]
Time [s]
15
10
5
0
−5
−10
−15
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

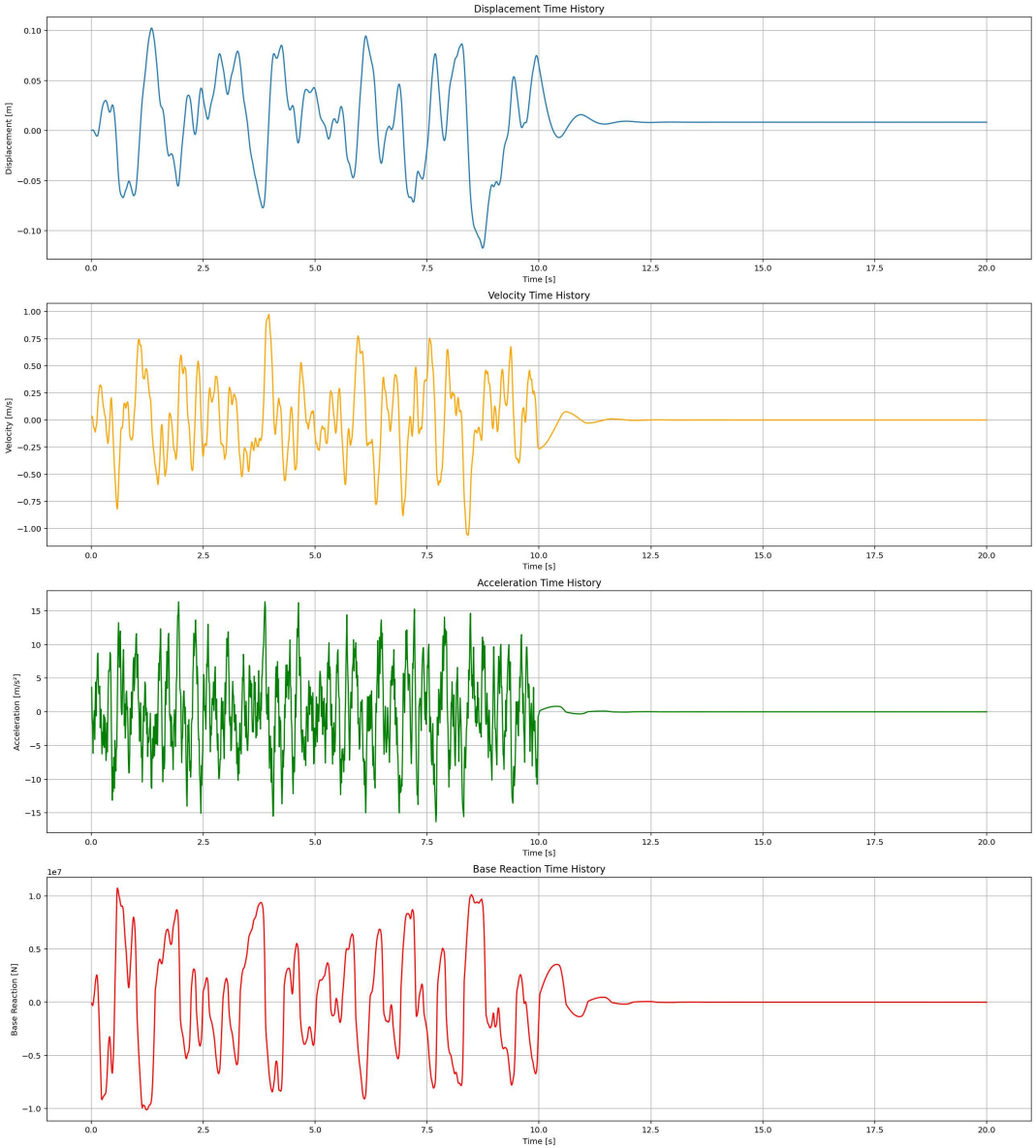

Time vs Base-reaction - MAX. ABS: 10720384.425908051
1e7
1.0
0.5
0.0
−0.5
−1.0
Base-reaction [N]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

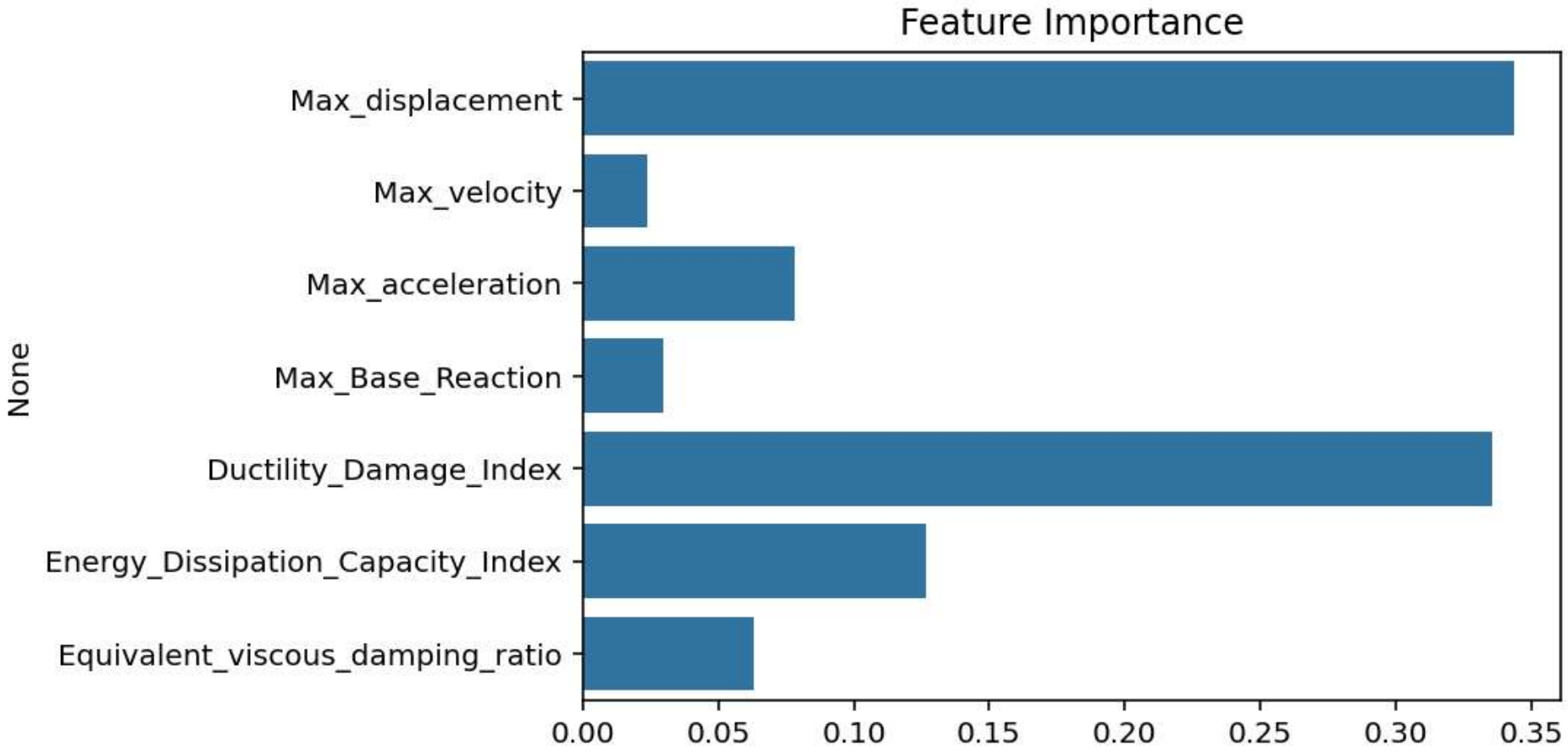

Base Reaction and Displacement scatter chart
1e7
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Base Reaction
Data
y = 44359701.60x + 5512106.94 - R^2 = 0.618
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Displacement

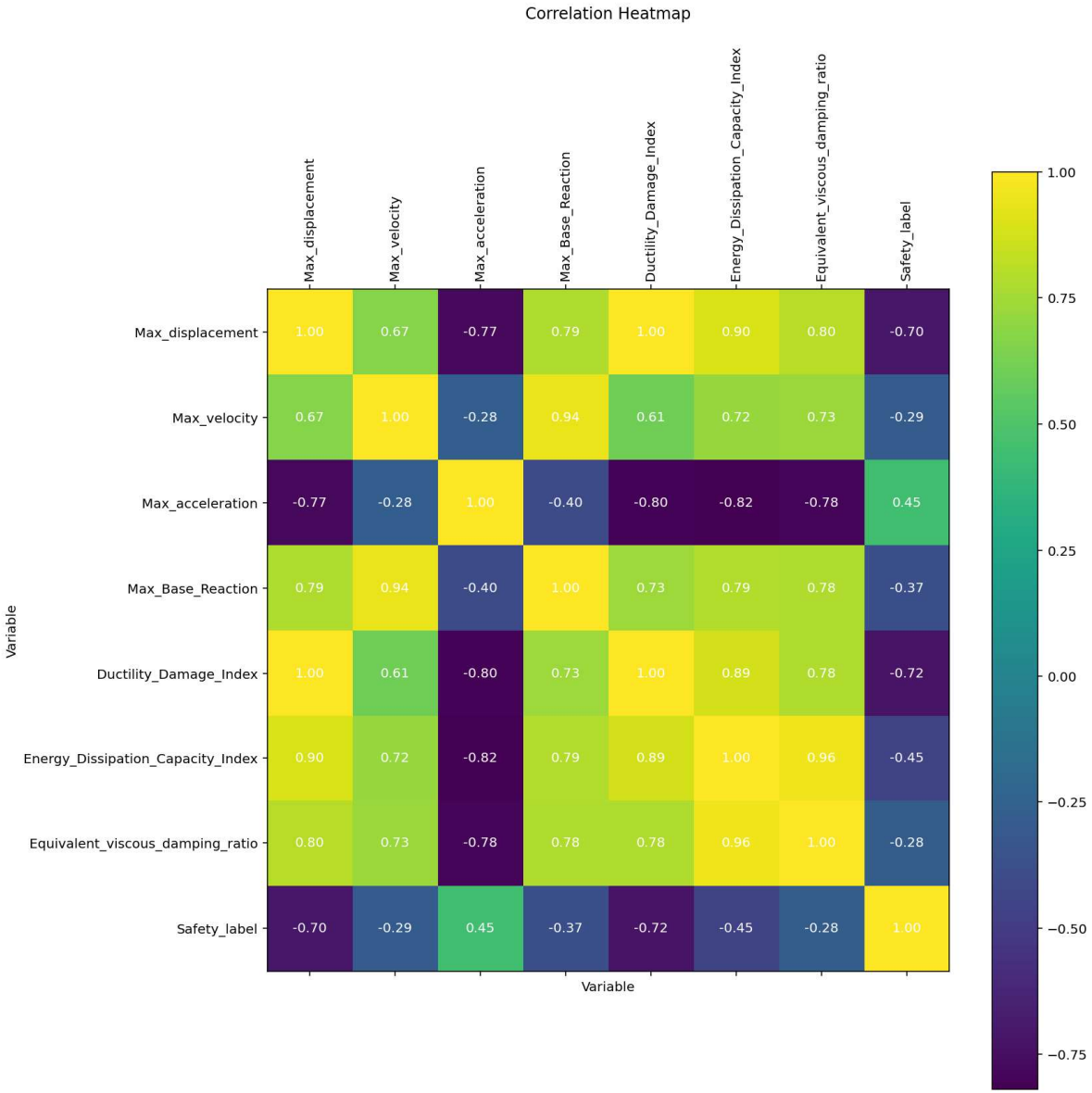

Base Reaction and Velocity scatter chart
1e7
Data
y = 9904747.30x + -213877.27 - R^2 = 0.890
Base Reaction
Velocity
0.0
0.2
0.4
0.6
0.8
1.0
0.2
0.4
0.6
0.8
1.0
1.2


Base Reaction and Acceleration scatter chart
1e7
Data
y = -119300.18x + 12311714.36 - R^2 = 0.156
Base Reaction
Acceleration
0.0
0.2
0.4
0.6
0.8
1.0
15
20
25
30
35
40
45
50


Structural Ductility Damage Index (%) and Displacement scatter chart
Data
y = 455.49x + -6.21 - R^2 = 0.993
Structural Ductility Damage Index (%)
Displacement
0
10
20
30
40
50
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14


Displacement Time History
Displacement [m]
Time [s]
Velocity Time History
Velocity [m/s]
Time [s]
Acceleration Time History
Acceleration [m/s²]
Time [s]
Base Reaction Time History
1e7
Base Reaction [N]
Time [s]


Feature Importance
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Energy_Dissipation_Capacity_Index
Equivalent_viscous_damping_ratio
None
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35


Correlation Heatmap
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Energy_Dissipation_Capacity_Index
Equivalent_viscous_damping_ratio
Safety_label
Variable
Variable
1.00 0.67 -0.77 0.79 1.00 0.90 0.80 -0.70
0.67 1.00 -0.28 0.94 0.61 0.72 0.73 -0.29
-0.77 -0.28 1.00 -0.40 -0.80 -0.82 -0.78 0.45
0.79 0.94 -0.40 1.00 0.73 0.79 0.78 -0.37
1.00 0.61 -0.80 0.73 1.00 0.89 0.78 -0.72
0.90 0.72 -0.82 0.79 0.89 1.00 0.96 -0.45
0.80 0.73 -0.78 0.78 0.78 0.96 1.00 -0.28
-0.70 -0.29 0.45 -0.37 -0.72 -0.45 -0.28 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75